

CHAPTER 11

Fuels and introduction to stoichiometry

In this chapter, you will learn how fuels are used to meet global energy needs and you will gain an appreciation of the chemistry that underpins decisions about the use of fuels. Combustion reactions are used to release useful heat energy from the chemical energy stored in fuels. You will explore how fuels vary in terms of the energy that they release when they are burnt.

You will also consider the environmental impact of using different types of fuels, including their carbon emissions as well as the other pollutants they release into the atmosphere. Current research being conducted into the production of renewable fuels and the potential for reducing the harmful impact of fossil fuels will also be discussed.

Science as a Human Endeavour

- There are differences in the energy output and carbon emissions of fossil fuels (including coal, oil, petroleum and natural gas) and biofuels (including biogas, biodiesel and bioethanol). These differences, together with social, economic, cultural and political values, determine how widely these fuels are used.

Science understanding

- fossil fuels (including coal, oil, petroleum and natural gas) and biofuels (including biogas, biodiesel and bioethanol) can be compared in terms of their energy output, suitability for purpose, and the nature of products of combustion
- the mole concept relates mass, moles and molar mass and, with the Law of Conservation of Mass, can be used to calculate the masses of reactants and products in a chemical reaction

WACE Chemistry ATAR Course Year 11 extracts © School Curriculum and Standards Authority, Government of Western Australia, 2014; reproduced by permission.

11.1 Types of fuels



FIGURE 11.1.1 Sugars, such as sucrose, are fuels for your body.



FIGURE 11.1.2 Petrol is just one type of fuel that is used each day to meet our energy needs.

Fuels provide us with energy. They are substances that have chemical energy stored within them. As you learnt in Chapter 10, all chemicals contain stored energy, which is called enthalpy. What makes a **fuel** special is that this stored chemical energy can be released relatively easily.

Sugar is an example of a common fuel (Figure 11.1.1). A cube of table sugar (sucrose) can provide your body with 82 kilojoules of energy. This is about 1% of your daily energy needs. If sucrose is burnt, this energy is released as heat. The combustion of 1 kilogram of sucrose releases sufficient energy to melt more than 5 kilograms of ice and then boil all the liquid water produced.

Although sugars provide energy for your body, you do not heat your home, power cars, or produce electricity by burning sugar! A range of other fuels such as wood, coal, oil, natural gas, LPG, ethanol and petrol (Figure 11.1.2) are used for these energy needs.

In this section, you will explore a range of different fuels and consider the chemical, economic and environmental factors that make them suitable for their purposes.

THE NEED FOR FUELS

A fuel is a substance with stored energy that can be released relatively easily for use as heat or power. Although this chapter will focus on fuels with stored chemical energy, the term ‘fuel’ is also applied to sources of nuclear energy, such as uranium.

The use of fuels by society can be considered from a number of points of view, including at a:

- local level, e.g. the type of fuel used to heat your house
- national level, e.g. whether Australia’s use of energy resources is sustainable
- global level, e.g. whether the use of fossil fuels (coal, oil and natural gas) is contributing to the enhanced greenhouse effect.

These are not separate issues. Choices made locally have regional and global effects. The decisions of global and national governments affect how and which fuels are used.

Units of energy

The international system of units (**SI units**) is a widely used system of measurement that specifies units for a range of quantities. The SI unit for energy is the joule, symbol J. As 1J of energy is a relatively small amount, it is common to see the following units in use:

- kilojoules, $1\text{ kJ} = 10^3\text{ J}$
- megajoules, $1\text{ MJ} = 10^6\text{ J}$
- gigajoules, $1\text{ GJ} = 10^9\text{ J}$
- terajoules, $1\text{ TJ} = 10^{12}\text{ J}$.

Use of energy in Australia and the world

World energy consumption is around 4×10^{20} joules per year. The United States consumes a quarter of the world’s energy. Australia consumes about one-hundredth of the world’s energy. But energy consumption per person in Australia is only just below that of the United States. Figure 11.1.3 shows the ways in which Australians use energy. As you can see, heating and transportation account for 87% of Australia’s total energy consumption.

In Australia and around the world, most of the energy used for heating, electricity generation and powering vehicles comes from fossil fuels. About 86% of Australian electricity is generated from these fuels, with 73% from coal and 13% from natural gas.

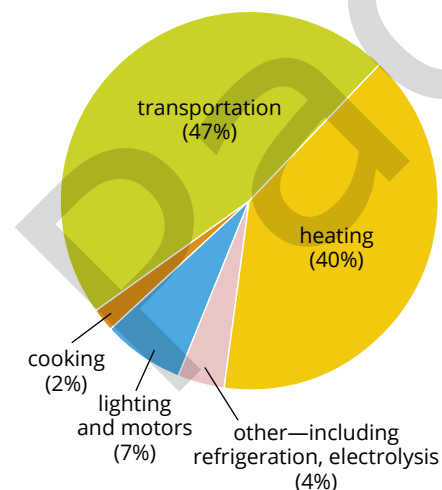


FIGURE 11.1.3 This pie chart shows where energy is used in Australia.

Coal-fired power stations are the dominant source of the world's electricity because they are often the cheapest form of generation. Electricity from coal-fired power stations is reliable and coal is very abundant.

About 14% of Australia's electricity comes from renewable energy sources. Hydroelectricity contributes 7% of total electricity and wind, biofuels and solar energy make up the other 7%.

Future energy needs

Burning wood was the dominant method of obtaining energy up to the middle of the 19th century. Wood supplies once seemed unlimited and, like fossil fuels today, satisfied most of the demands of the time.

Fossil fuels (coal, oil and gas) now provide nearly 90% of the world's energy needs. As members of a society that is heavily dependent on fossil fuels as a source of energy, we can at times find it hard to imagine obtaining energy from elsewhere.

The world first became aware that fossil fuels are a finite energy reserve during the 'oil crisis' of the early 1970s. Several Middle Eastern oil exporters restricted production for political reasons. This dramatically increased the cost of crude oil and caused huge increases in the price of petrol around the world.

Given the limited reserves and concerns about the link between fossil fuels and climate change, there is considerable interest in identifying and developing new energy sources. Many countries are already considering alternative sources of energy. The development of alternative sources for large-scale energy production is not a simple task as these replacement sources need to be reliable and cost effective. There is also a growing pressure for new sources to be more sustainable and renewable than the currently used fossil fuels. These factors are due to the growing worries about the effect of burning fossil fuels on global warming and the limited supplies of fossil fuels. Figure 11.1.4 shows the increase in world energy production from different sources.

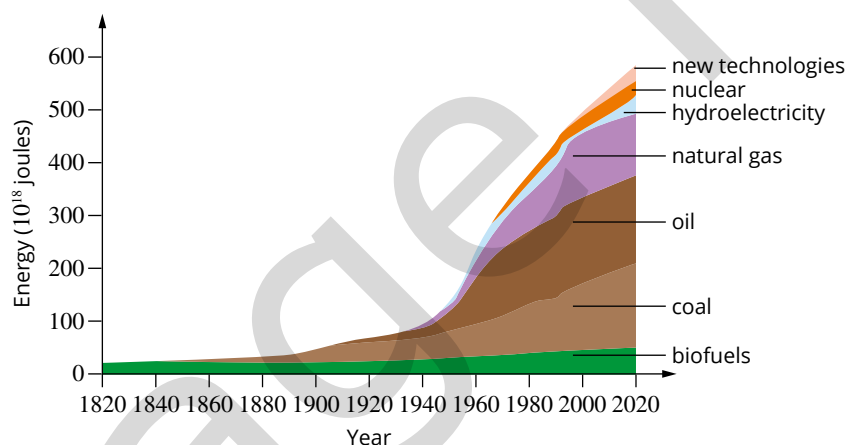


FIGURE 11.1.4 This graph shows the increase in world energy production.

FOSSIL FUELS

Non-renewable resources are those that are used faster than they can be replaced. Coal, oil and natural gas are non-renewable fuels. Reserves of fossil fuels are limited and they could eventually be exhausted.

There is an ongoing conflict between using non-renewable resources and **sustainability**. Sustainability is the combination of meeting the long-term ongoing needs of society, whilst still reaching the immediate and short-term demands. This concept incorporates the care of the physical environment as well as the economic and social wellbeing of future generations.

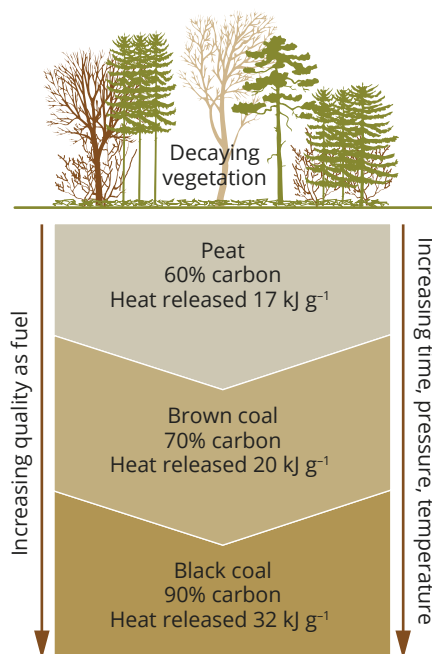


FIGURE 11.1.5 Steps in the formation of coal. Values of the carbon content and heat released upon combustion are for dried coal.



FIGURE 11.1.6 In Australia, brown coal is mined in the Latrobe Valley in Victoria.

Formation of fossil fuels

Coal, oil and natural gas were formed from ancient plants, animals and microorganisms. Buried under tonnes of mud, sand and rock, this once biological material has undergone complex changes to become the fossil fuels used by societies today. The organic matter still retains some of the chemical energy the plants originally accumulated by carrying out **photosynthesis**. Chemical energy in fossil fuels can be considered to be trapped solar energy.

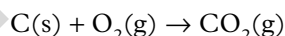
Fossil fuel formation occurs over millions of years. This is why these fuels are considered non-renewable. Once reserves of the fossil fuels have been used, they will not be replaced in the foreseeable future.

Coal

As wood and other plant material turn into coal, gradual chemical changes occur. Wood is about 50% carbon and as it is converted into coal, the carbon content increases and the proportion of hydrogen and oxygen decreases. The wood progressively becomes peat, brown coal and then black coal (Figure 11.1.5). Coal is a mixture of large molecules made from carbon, hydrogen, nitrogen, sulfur and other elements.

The amount of water in coal decreases as these changes occur. When coal is burned, the energy released causes the water to vaporise, reducing the net amount of heat released. Black coal, which contains the least water and therefore the highest percentage of carbon, is a better fuel than brown coal or peat.

Although black coal is usually buried further underground than brown coal, its higher heat value often makes it economical to mine. Black coal is mined in the Collie region in Western Australia. The coal produced there is highly competitive when compared with natural gas and other coal deposits, as it has a low sulfur content and generates low levels of noxious gases during combustion. This black coal is used for domestic power generation as well as being exported overseas. Over three-quarters of Australia's electricity is sourced from coal. The reaction that occurs when coal burns can be written as:



This equation is a simplification of the electricity creating process, as coal is not pure carbon. Significant quantities of ash are formed, as well as water vapour and sulfur dioxide.

Large brown coal deposits are located in the Latrobe Valley in Victoria (Figure 11.1.6). The power stations located next to these open-cut mines burn brown coal to generate electricity. Australia is the fifth largest producer of coal and the second largest exporter of coal in the world.

CHEMFILE

Forming crude oil

The main deposits of crude oil were formed from small marine animals (zooplankton) and plants (phytoplankton) that lived up to 1 billion years ago. Some crude oil deposits are estimated to be even older, as much as 3–4 billion years old. If a deposit of crude oil was trapped beneath a layer of impermeable rock, then a layer of natural gas would also form.

The first deposits of crude oil were discovered at the end of the 19th century in the United States. Today, the largest crude oil deposits are in Russia, Iran, Iraq and Saudi Arabia. New crude oil deposits

are still being found throughout the world. The oldest deposits found so far are in Venezuela, where the oil is estimated to be almost 4 billion years old. However, only about 10% of the oil discovered is profitable to extract.

Permeable rocks contain tiny spaces through which liquid substances can move. Crude oil has a lower density than water, so oil migrates upwards through permeable rocks over time. Large deposits of oil are formed when portions of this migrating oil become trapped under impermeable rocks. To extract the crude oil, drilling into the impermeable rock has to take place (Figure 11.1.7). In most cases, the oil flows up by itself under

high pressure that has gradually built up from when the oil was formed. As the extraction continues, the overall pressure drops and pumps are needed to extract the remaining deposit.



FIGURE 11.1.7 Operating drill during oil and gas exploration

Fuels from crude oil

Crude oil (also called petroleum) is a mixture of hydrocarbon molecules that are mostly alkanes, which were introduced in Chapter 8. Crude oil itself is of no use as a fuel, but its components can be made into many important fuels such as petrol, kerosene, diesel and liquefied petroleum gas (LPG).

The relative amounts of different alkanes in crude oil vary with the deposit. For example, oil from Bass Strait and the Carnarvon Basin, Western Australia, contains relatively few of the larger molecules needed to form lubricants and bitumen.

Crude oil is separated into a range of fractions by fractional distillation. Fractional distillation does not produce pure substances. Each fraction is still a mixture of hydrocarbon compounds. These fractions can be used as fuels, or treated further to produce more specific products through chemical processes.

i Alkanes are hydrocarbons with the general formula C_nH_{2n+2} . Alkanes are commonly found in crude oil.

CHEMISTRY IN ACTION

Fractional distillation of crude oil

Fractional distillation uses heat to separate a mixture into a number of different parts or fractions. A number of different temperatures are produced because of the nature of the column in which the crude oil is placed. The temperature of the tower decreases gradually with increasing height. Within the tower are horizontal trays, each containing hundreds of bubble caps. Bubble caps impede (stop) the upwards movement of gases (Figure 11.1.8). As the vapour rises, it forces the caps up and it bubbles through condensed liquid in the trays. Those substances in the vapour that have boiling points almost equal to the temperature of the liquid in the trays condense and are collected. Consequently, fractions

collected from trays higher in the tower will have lower boiling points.

The boiling point of a molecular compound depends on the strength of its intermolecular forces, which are stronger with increasing molecular mass. As a result, each fraction consists of alkanes within a specific mass range. Lighter alkanes (lower boiling point) condense near the top of the tower, whereas heavier alkanes (higher boiling point) condense near the bottom. The composition and boiling range of each fraction are summarised in Figure 11.1.8. For example, the petrol fraction that boils (and condenses) between 100°C and 250°C consists of alkanes containing 8–12 carbon atoms; that is, C_8H_{18} to $C_{12}H_{26}$.

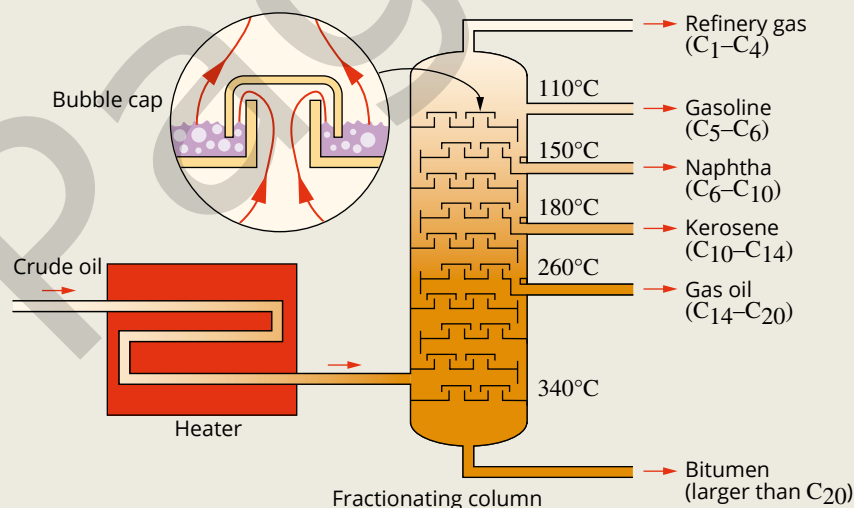


FIGURE 11.1.8 Fractional distillation of oil, showing the bubble caps that stop the upward movement of gases

Australia's contains relatively small amounts of oil reserves compared to the reserves in the Middle East. Figure 11.1.9 shows the current locations of Australian oil, gas and coal fields. These reserves are associated with giant offshore gas fields near the northern Western Australian coast, and in reserves in outback South Australia and in Bass Strait. Western Australia currently produces 71% of Australia's crude oil and this value is anticipated to rise due to new supply sources that have been found. However Australia's production is set to decline gradually and will likely be exhausted later this century. Australia already imports over 90% of the crude oil it uses. Importation of large amounts of oil has a significant impact on Australia's economy.



Natural gas

Natural gas is another fossil fuel found in deposits in the Earth's crust. It is a popular fuel for home heating and cooking. Natural gas is mainly composed of methane (CH_4) together with small amounts of other hydrocarbons such as ethane (C_2H_6) and propane (C_3H_8). Water, sulfur, carbon dioxide and nitrogen may also be present in natural gas.

Natural gas can be found:

- in gas reservoirs trapped between layers of rocks
- as a component of petroleum deposits
- in coal deposits where it is bonded to the surface of the coal. Coal seams usually contain water and the pressure of the water can keep the gas adsorbed to the coal surface. Natural gas found this way is known as coal seam gas or CSG. It is a major component of the energy supplies of Queensland
- trapped in shale rock, where it is referred to as shale gas. Shale gas is mined in many parts of the United States.

Natural gas is accessed by drilling as with crude oil; drilling causes the natural gas to flow to the surface (Figure 11.1.10).

Propane and butane gases can be separated from natural gas by fractional distillation. Propane and butane become liquids under pressure and are sold as **liquefied petroleum gas** (LPG). The natural gas remaining after the removal of propane and butane is used widely as a fuel for home heating and cooking.

LPG has applications as a motor fuel and in home gas bottles. As world oil prices rise, natural gas could increasingly replace petrol in cars and other vehicles. LPG is already used to power all many of Transperth's buses.

Fracking

The extraction of natural gas from coal or shale deposits usually involves a process called **fracking**. Under pressure, the natural gas is adsorbed on the surface of the coal or shale. Fracking is used to fracture the rock or coal to release the natural gas.

Fracking begins with drilling a well into the deposit to access the trapped gas. The well is encased in steel and concrete to prevent leakage into local water supplies. Fracking fluid is then pumped down the well at extremely high pressures. This high-pressure fluid fractures the surrounding rock or coal, creating fissures through which gas can flow. This process is shown in Figure 11.1.11.

Materials used for fracking include sand, water and other chemicals. There are concerns about the potential impact of this process on the local environment and underground water supplies.

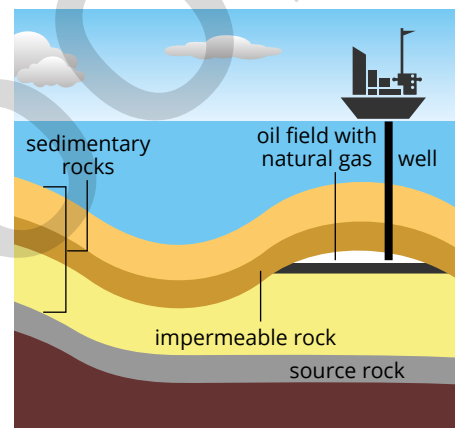


FIGURE 11.1.10 Natural gas deposits are often found trapped above crude oil. Once a well is sunk into the deposit, the natural gas flows to the surface.

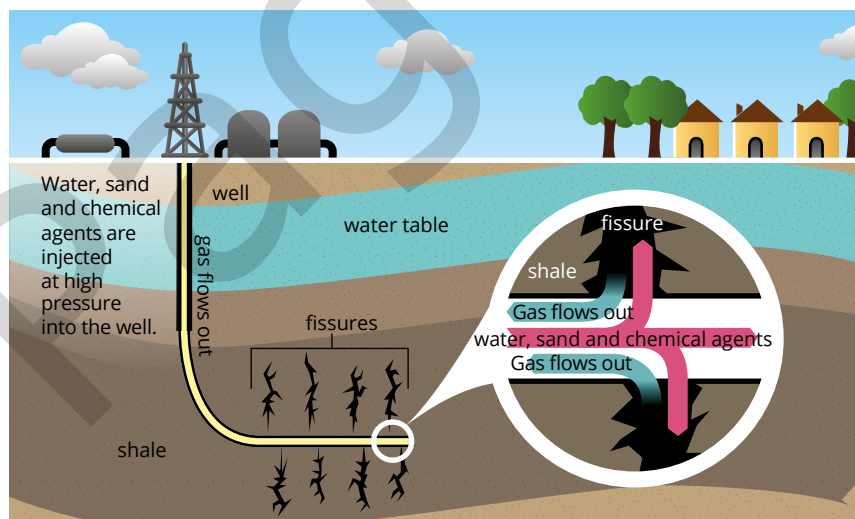


FIGURE 11.1.11 The fracking process: sand, water and other chemicals are injected into the deposit at high pressure to free the gas from the coal or shale.

Debate surrounding coal seam gas

Coal seam gas (CSG) is a natural gas that is extracted from underground coal seams, where it is trapped in pores in the coal. Almost 30% of Australia's natural gas reserves come from coal seam gas. Reserves of CSG are found in multiple areas of Western Australia, as seen in Figure 11.1.12.

Various chemicals have been used as fracking fluids. Fracking fluids increase the permeability of the rock and therefore flow of gas to the surface. Fluids such as benzene, toluene, xylene and ethylbenzene were once commonly used. The use of these fluids has been banned in both New South Wales and Queensland because of concern over their effect on the environment. For example, these potentially carcinogenic compounds may escape and contaminate groundwater.

Water is now commonly used as a fracking fluid. Generally, large amounts of water are not available at the fracking site, so water needs to be transported in, which can have significant economic and environmental costs.

Since 2005, 15 exploration wells have been drilled in WA to search for coal seam gas. The Environmental Protection Agency has not placed a ban on fracking, and instead assesses every possible project on a case-by-case basis. Currently there are still large offshore reserves of oil and gas that WA relies on, however it is predicted that coal seam gas will become industrialised in the near future, despite the concerns of many regional community members.

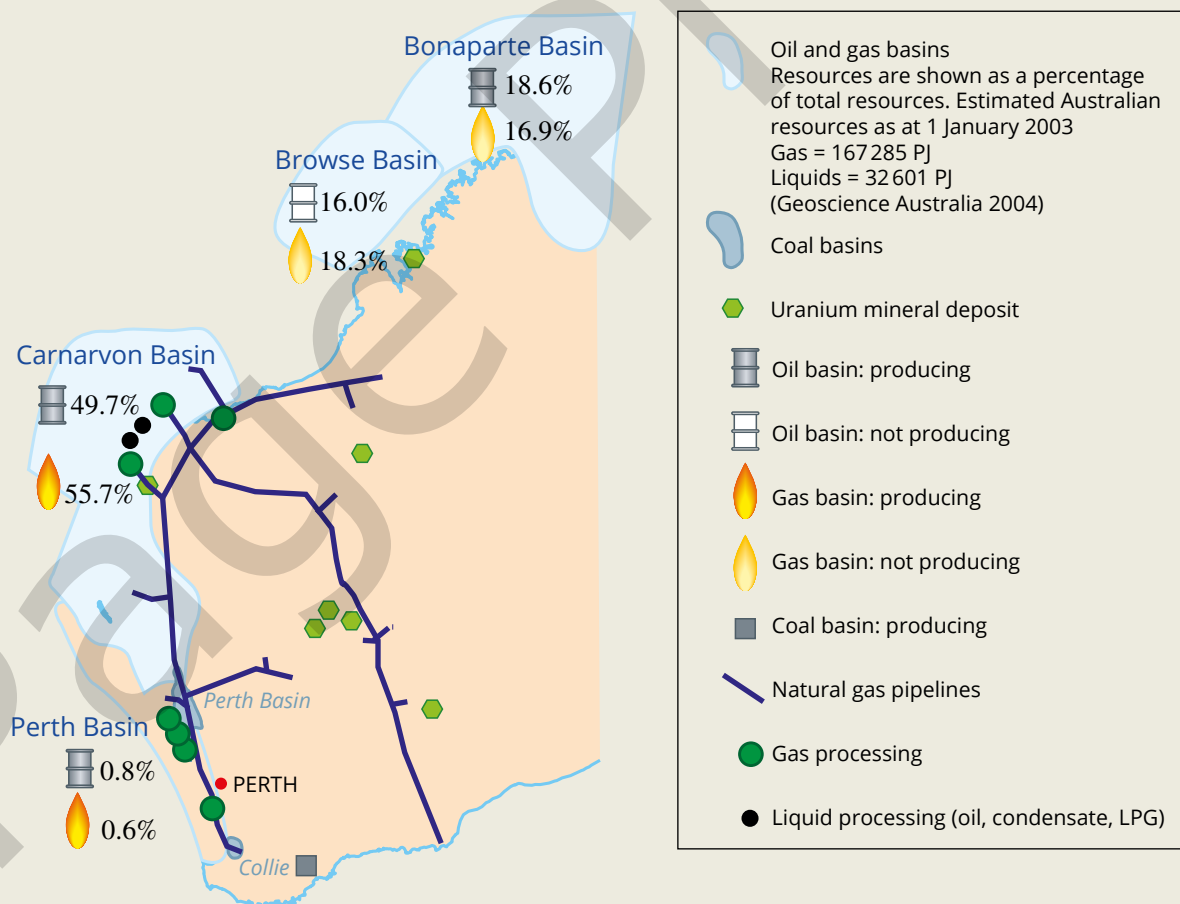


FIGURE 11.1.12 Location of gas basins in Western Australia. The basins the mainland can be accessed by fracking.

BIOFUELS

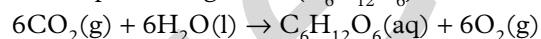
Ideally, new sources of energy will be clean and **renewable**. Renewable energy is energy that can be obtained from natural resources that can be constantly replenished.

Biochemical fuels (or **biofuels**) are fuels derived from plant materials such as grains (maize, wheat, barley or sorghum), sugar cane (Figure 11.1.13) and vegetable waste, and vegetable oils. The three common biofuels are **biogas**, **bioethanol** and **biodiesel**. They are used alone or blended with fossil fuels such as petrol and diesel.



FIGURE 11.1.13 Harvesting sugar cane in Queensland. Sugar cane can be a source of the raw materials for the production of bioethanol.

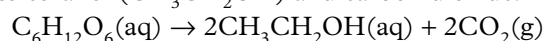
As well as being renewable, biofuels are not considered to contribute significantly to an increase in the atmospheric carbon dioxide levels. The plant materials used in the generation of biofuels are produced by photosynthesis, which removes carbon dioxide from the atmosphere and produces glucose ($C_6H_{12}O_6$) in the following reaction:



The plants convert the glucose into cellulose and starch. Although carbon dioxide is released back into the atmosphere when the biofuel is burnt, the net impact should be less than for fossil fuels. The use of biochemical fuels can therefore be described as **carbon neutral**, although there will be some carbon emissions from the manufacture and distribution of biochemical fuels. Therefore, there will always be some net emissions associated with their use.

Bioethanol

For thousands of years, humans have employed biological catalysts (**enzymes**) from yeasts to convert starches and sugars to ethanol. Enzymes catalyse the breakdown of the starch in grain crops (such as barley and wheat) to glucose. **Fermentation** is a process which uses other enzymes from yeasts to convert glucose and other small sugar molecules to ethanol (CH_3CH_2OH) and carbon dioxide:



Fermentation is an anaerobic process as it takes place in the absence of oxygen.

Bioethanol is used extensively in Australia, including E10 petrol which contains 10% ethanol. This mix can be used by most modern car engines and its use reduces the consumption of petrol derived from crude oil whilst also reducing the total amount of carbon dioxide that is released into the atmosphere. In addition, the use of E10 fuel reduces the quantity of particles such as unburnt carbon and gases like the oxides of nitrogen that are emitted by exhausts and contribute to air pollution.

Biogas

Biogas is released in the breakdown of organic waste by **anaerobic** bacteria. These bacteria decompose the complex molecules contained in substances such as carbohydrates and proteins into the simple molecular compounds carbon dioxide and methane. A digester (Figure 11.1.14) is a large tank filled with the anaerobic bacteria that digest the complex molecules to form biogas.

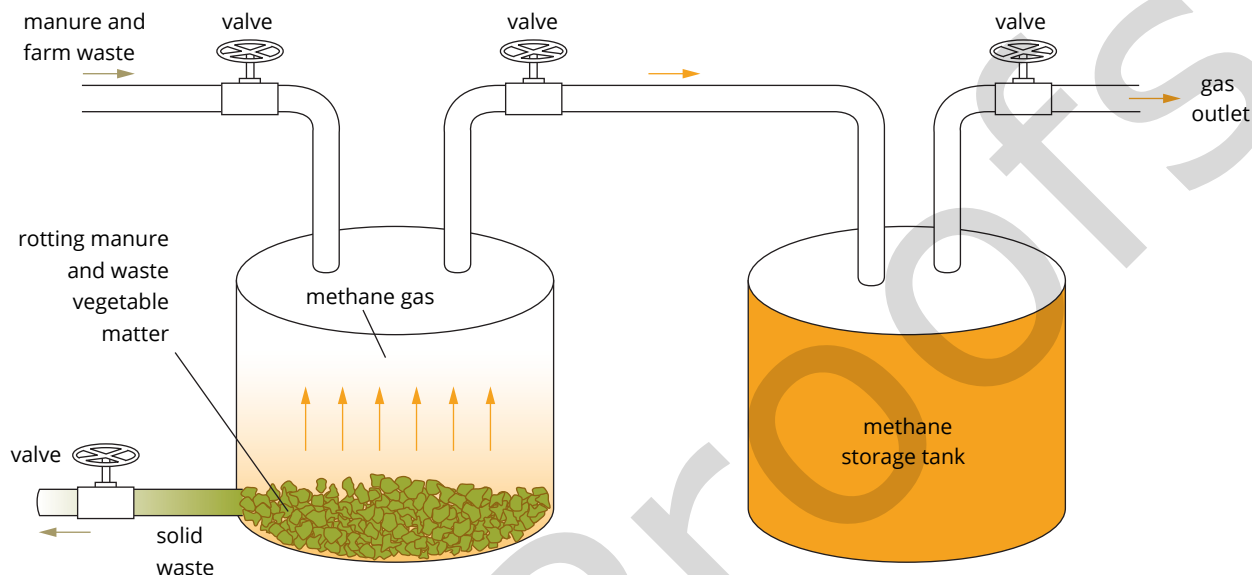


FIGURE 11.1.14 A digester is used in the production of biogas.



FIGURE 11.1.15 Pipes buried in this rubbish tip collect biogas.

A range of materials, including rotting rubbish (such as that seen in Figure 11.1.15) and decomposing plant material, can be used to produce biogas.

The composition of biogas depends on the original material from which it is obtained and the method of decomposition. The typical composition of a sample of biogas is shown in Table 11.1.1.

TABLE 11.1.1 Typical percentage composition of different molecules found in biogas

Gas	Formula	Percentage composition (by volume)
methane	CH_4	60
carbon dioxide	CO_2	32
nitrogen	N_2	4.5
hydrogen sulfide	H_2S	2
oxygen	O_2	1
hydrogen	H_2	0.5

As you can see from Table 11.1.1, biogas consists mainly of methane and carbon dioxide. Biogas can be used for heating and to power homes and farms. There are more than 7 million biogas generators in China. Biogas generators are particularly suited to farms, as the waste from a biogas generator makes a rich fertiliser.

In the future, it is likely more energy will be obtained from biogas generated at sewage works, chicken farms, piggeries and food-processing plants. Your local rubbish tip also has the potential to supply biogas. The gas can be used directly for small-scale heating or to generate electricity.

Biodiesel

Biodiesel is a mixture of organic compounds called esters, which are produced by a chemical reaction between vegetable oils or animal fats and an alcohol, most commonly methanol (CH_3OH).

The usual raw material for the production of biodiesel is vegetable oil from sources such as soyabean, canola or palm oil. Recycled vegetable oil or animal fats can also be used. The structure of a typical biodiesel molecule is shown in Figure 11.1.16.

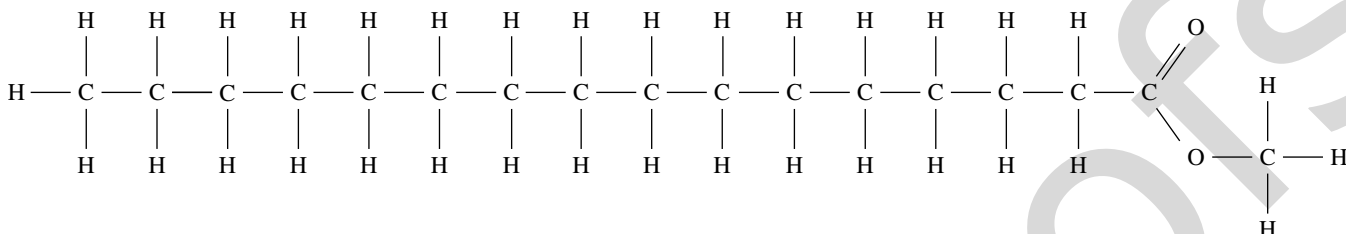


FIGURE 11.1.16 Structural formula of a typical biodiesel molecule

Petrodiesel is a hydrocarbon fraction obtained from crude oil. The chemical and physical properties of biodiesel are very similar to those of petrodiesel, so it is possible to run diesel vehicles on 100% biodiesel fuel.

A diesel engine is a form of internal combustion engine that does not require a spark to ignite the fuel in the engine cylinder. The high energy content of diesel makes it an excellent fuel for transportation and equipment designed to do heavy work. Therefore diesel engines are favoured in situations requiring high fuel efficiencies and for heavy haulage vehicles.

Diesel engines have a 20–40% better fuel economy than petrol engines because:

- diesel engines are typically more energy efficient than petrol engines
- diesel fuel has a higher density than petrol, so although the energy content of diesel and petrol, measured in kJ g^{-1} , is similar, diesel fuel yields more energy per litre.

Biodiesel is biodegradable, non-toxic and produces fewer pollutants in vehicle emissions. Like ethanol, the use of biodiesel does not add to the overall amount of carbon dioxide already present in the atmosphere as the creation of the raw materials that make biodiesel take carbon dioxide from the atmosphere.

CHEMFILE

Rudolf Diesel

Rudolf Diesel (1858–1913) (Figure 11.1.17) was a German engineer who invented the ‘oil engine’ that was named after him. His prototype engine first operated in 1893 and was powered by peanut oil. Diesel was well aware of the potential value of an engine that could run on renewable fuel. He demonstrated his engine at the World Exhibition in Paris in 1900 and was awarded the Grand Prix—the highest prize.

Just before his death in 1913, Diesel stated: ‘The diesel engine can be fed with vegetable oils and would help considerably in the development of agriculture of the countries which use it.’

Diesel also predicted that: ‘The use of vegetable oils for engine fuels may seem insignificant today. But such oils may become, in course of time, as important as petroleum and the coal tar products of the present time.’

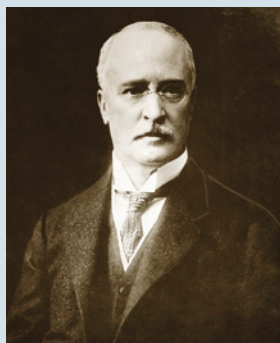


FIGURE 11.1.17 Rudolf Diesel, inventor of the diesel engine

11.1 Review

SUMMARY

- A fuel is a substance with stored energy that can be released relatively easily.
- A fuel is considered to be non-renewable if it cannot be replenished at the rate at which it is consumed. Fossil fuels such as coal, oil and natural gas are non-renewable.
- Fossil fuels are produced over millions of years by the breakdown of biomass at high temperatures and pressures underground.
- Australia has large reserves of coal and natural gas.
- A fuel is considered to be renewable if it can be replenished at the rate at which it is consumed. Biofuels such as biogas, bioethanol and biodiesel are renewable.
- Biogas is formed by the anaerobic breakdown of organic waste.
- Bioethanol can be produced by fermentation of starches and sugars.
- Biodiesel is produced in a reaction between a vegetable oil or an animal fat and a small alcohol molecule such as methanol.
- Some of the non-renewable and renewable fuels in use in Australia are listed in Table 11.1.2.
- Diesel engines are used in many forms of transport and in heavy-duty equipment.

TABLE 11.1.2 Types of renewable and non-renewable fuels in use in Australia

Non-renewable fuels	Renewable fuels
coal	bioethanol
oil	biogas
liquefied petroleum gas (LPG)	biodiesel
natural gas	
coal seam gas (CSG)	

KEY QUESTIONS

- 1 What is the difference between a renewable and non-renewable fuel?
- 2 Give an example of a renewable fuel source and a non-renewable fuel source used in Australia.
- 3 In Australia, which resource is likely to last longer before it is depleted: coal, oil or natural gas? Explain your answer.
- 4 Wood from forests is a renewable resource that supplied global energy needs for thousands of years.
 - a Why is wood no longer sustainable as the major energy source for today's society?
 - b Is it possible to have a non-renewable and sustainable energy source? Explain.
- 5 Why is it necessary to treat crude oil by fractional distillation?
- 6 Why are CO₂ emissions from the use of biodiesel not considered as problematic as those produced from the use of petrodiesel?

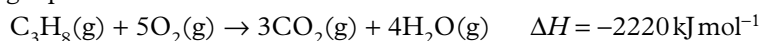
11.2 Combustion reactions

Fuels contain stored chemical energy that can be harnessed to perform useful functions. The release of this energy occurs when the fuel is burnt in the presence of oxygen, with this process being known as **combustion**. The heat energy released during combustion provides heat for warmth and cooking, as well as acting as the source of electrical energy and mechanical energy for transport.

In this section, you will look at different fuels used to produce electricity or power vehicles and compare the environmental impact of these fuels.

COMBUSTION AS A CHEMICAL PROCESS

Combustion reactions are exothermic reactions in which the reactant combines with oxygen to produce oxides. The combustion of a hydrocarbon produces carbon dioxide and water, provided there is enough oxygen present. An example is the combustion of propane (C_3H_8), a major component of LPG, shown in the following equation:



Note the following important features of this thermochemical equation.

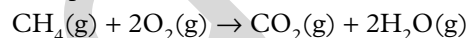
- No atoms are created or destroyed in the reaction, so the numbers of atoms of each element are balanced on the two sides of the equation.
- The balanced equation tells you the number of molecules of oxygen that are required for the combustion of each molecule of propane.
- The enthalpy change for the reaction is negative, indicating that this is an exothermic reaction.
- The enthalpy change is given in kJ mol^{-1} . It tells you the energy released, in kJ, according to the coefficients given in the equation. (In this case, 1 mole of propane is combusted.)
- State symbols show the state of each reactant and product. (In combustion reactions, water is released as water vapour—a gas.)

Combustion of hydrocarbons

Combustion reactions can be described as complete or incomplete. The difference between the two is due to the amount of oxygen available to react with the fuel.

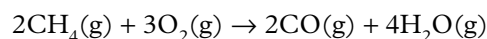
Complete combustion occurs when oxygen is plentiful. The only products are carbon dioxide and water.

An example is the complete combustion of methane:



When the oxygen supply is limited, **incomplete combustion** occurs. As less oxygen is available, not all of the carbon can be converted into carbon dioxide. Carbon monoxide and/or carbon are produced instead. The hydrocarbon burns with a yellow, smoky or sooty flame, due to the presence of glowing carbon particles. Figure 11.2.1 shows the appearance of the different flames of a Bunsen burner due to incomplete and complete combustion.

The equation for the incomplete combustion of methane to form carbon monoxide is:



i The complete combustion of hydrocarbons occurs when there is sufficient oxygen for the fuel to burn. The products of complete combustion are carbon dioxide and water. When oxygen is not plentiful, incomplete combustion occurs. The products of incomplete combustion are carbon monoxide and/or carbon and water.



FIGURE 11.2.1 The yellow flame of a Bunsen burner is due to incomplete combustion and produces carbon as a product. The blue flame is a hotter flame that occurs when the collar hole is open and more oxygen is allowed into the reaction. Complete combustion can then occur.

CHEMFILE

Carbon monoxide poisoning

Carbon monoxide is a highly poisonous gas. It combines readily with haemoglobin, the oxygen carrier in blood. When attached to carbon monoxide, haemoglobin cannot transport oxygen around the body, which leads to oxygen starvation of tissues.

Even at concentrations as low as 10 parts per million (ppm), carbon monoxide can cause drowsiness, dizziness and headaches. At about 200 ppm, carbon monoxide can lead to death. The average carbon monoxide concentration in large cities, mostly due to incomplete combustion of fuels in cars (Figure 11.2.2), is now 7 ppm, but it can be as high as 120 ppm at busy intersections in heavy traffic.



FIGURE 11.2.2 Car exhaust gases can contain high levels of carbon monoxide as a result of incomplete combustion of fuels.

WRITING EQUATIONS FOR THE COMPLETE COMBUSTION OF FUELS

Chemical equations tell us a lot about reactions and therefore it is important that we write them correctly. Writing equations for the complete combustion of fuels containing carbon and hydrogen is relatively straightforward, because the products are always carbon dioxide and water.

Perhaps the most important of all combustion reactions involving fuels are those that occur when petrol is burnt. Petrol is a mixture of hydrocarbons, including octane.

The combustion reactions of octane (C_8H_{18}) and the other hydrocarbons in petrol power the internal combustion engines in most of Australia's 17.6 million motor vehicles.

Worked example 11.2.1

WRITING EQUATIONS FOR COMPLETE COMBUSTION OF HYDROCARBON FUELS

Write the equation, including state symbols, for the complete combustion of butane (C_4H_{10}).	
Thinking	Working
Add oxygen as a reactant and carbon dioxide and water as the products.	$\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
Balance carbon and hydrogen atoms, based on the formula of the hydrocarbon.	$\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow 4\text{CO}_2 + 5\text{H}_2\text{O}$
Find the total number of oxygen atoms on the product side.	Total O $= (4 \times 2) + 5$ $= 13$
If this is an odd number, multiply all of the coefficients in the equation by two, except for the coefficient of oxygen.	$2\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow 8\text{CO}_2 + 10\text{H}_2\text{O}$
Balance oxygen by adding the appropriate coefficient to the reactant side of the equation.	$2\text{C}_4\text{H}_{10} + 13\text{O}_2 \rightarrow 8\text{CO}_2 + 10\text{H}_2\text{O}$
Add state symbols.	$2\text{C}_4\text{H}_{10}(\text{g}) + 13\text{O}_2(\text{g}) \rightarrow 8\text{CO}_2(\text{g}) + 10\text{H}_2\text{O}(\text{g})$

Worked example: Try yourself 11.2.1

WRITING EQUATIONS FOR COMPLETE COMBUSTION OF HYDROCARBON FUELS

Write the equation, including state symbols, for the complete combustion of hexane (C_6H_{14}).

A similar series of steps can also be used to write the combustion equations for other carbon-based fuels that contain oxygen; for example, alcohols.

Worked example 11.2.2

WRITING EQUATIONS FOR COMBUSTION REACTIONS OF ALCOHOLS

Write the equation, including state symbols, for the complete combustion of liquid ethanol ($\text{C}_2\text{H}_5\text{OH}$).	
Thinking	Working
Add oxygen as a reactant and carbon dioxide and water as the products.	$\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
Balance carbon and hydrogen atoms, based on the formula of the alcohol.	$\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$
Find the total number of oxygen atoms on the product side. Then subtract the one oxygen atom in the alcohol molecule from the total number of oxygen atoms on the product side.	Total O on product side $= (2 \times 2) + 3$ $= 7$ Total O on product side – 1 in alcohol $= 7 - 1 = 6$
If this is an odd number, multiply all the coefficients in the equation by two, except for the coefficient of oxygen.	$\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$
Balance oxygen by adding the appropriate coefficient to the reactant side of the equation.	$\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$
Add state symbols.	$\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{g})$

Worked example: Try yourself 11.2.2

WRITING EQUATIONS FOR COMBUSTION REACTIONS OF ALCOHOLS

Write the equation, including state symbols, for the complete combustion of liquid methanol (CH_3OH).

WRITING EQUATIONS FOR INCOMPLETE COMBUSTION OF FUELS

When the supply of oxygen is insufficient the combustion of fuels might not be complete. These incomplete combustion reactions can also be represented by equations. In general, for the incomplete combustion of hydrocarbons, as well as carbon-based fuels that contain oxygen, the products are carbon monoxide and/or carbon and water.

Worked example 11.2.3

WRITING EQUATIONS FOR INCOMPLETE COMBUSTION OF FUELS

Write an equation, including state symbols, for the incomplete combustion of ethane gas (C ₂ H ₆) to form carbon monoxide and water vapour.	
Thinking	Working
Add oxygen as a reactant and carbon monoxide and water as the products.	$\text{C}_2\text{H}_6 + \text{O}_2 \rightarrow \text{CO} + \text{H}_2\text{O}$
Balance the carbon and hydrogen atoms, based on the formula of the hydrocarbon.	$\text{C}_2\text{H}_6 + \text{O}_2 \rightarrow 2\text{CO} + 3\text{H}_2\text{O}$
Balance oxygen by adding the appropriate coefficient to the reactant side of the equation.	$\text{C}_2\text{H}_6 + \frac{5}{2}\text{O}_2 \rightarrow 2\text{CO} + 3\text{H}_2\text{O}$
If oxygen gas has a coefficient that is half of a whole number, multiply all of the coefficients in the equation by two.	$2\text{C}_2\text{H}_6 + 5\text{O}_2 \rightarrow 4\text{CO} + 6\text{H}_2\text{O}$
Add state symbols.	$2\text{C}_2\text{H}_6(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{CO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$

Worked example: Try yourself 11.2.3

WRITING EQUATIONS FOR INCOMPLETE COMBUSTION OF FUELS

Write an equation, including state symbols, for the incomplete combustion of liquid methanol (CH₃OH) to form carbon monoxide and water vapour.

ENVIRONMENTAL IMPACT

A discussion of the environmental impact of fuels needs to consider both the impact of emissions from the combustion of the fuel, and the impact on the environment of obtaining the fuel in the first place.

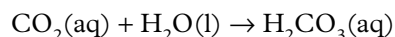
Emissions from fuel combustion

Carbon dioxide

As you have seen, carbon dioxide is produced when carbon based fuels, like wood, coal, gas and ethanol undergo combustion. Carbon dioxide is vital to life on earth. Through the process of photosynthesis, in the presence of chlorophyll (a plant pigment) and sunlight, green plants convert carbon dioxide and water to glucose and oxygen. This glucose produced is the source of energy that all aerobic (oxygen-using) life forms utilise for energy.

Because large quantities of fuel are burnt every day to meet society's energy needs, the level of carbon dioxide production is high. Carbon dioxide is not a toxic gas at normal concentrations in our atmosphere, however it can have many negative impacts on our environment as a whole.

When carbon dioxide dissolves in water, the majority of it will remain as unreacted CO₂(aq). However a small amount will react with water to form carbonic acid, H₂CO₃:



This will cause a slight acidity to large water bodies, and as such is the reason that rainwater is slightly acidic. An increase in atmospheric carbon dioxide concentration will cause an increase in the acidity of the oceans. This process is called **acidification**.

Due to its molecular structure, carbon dioxide can absorb and re-emit infrared radiation, also known as heat. This feature makes it a **greenhouse gas**.

Energy from the Sun heats the surface of the Earth. The Earth in turn radiates energy back towards space but greenhouse gases in the atmosphere absorb and re-radiate the energy in a process known as the **greenhouse effect**. The higher the concentration of a greenhouse gas, the more energy is trapped.

The greenhouse effect occurs naturally due to the gases present in the atmosphere. It has enabled the Earth to maintain its average temperature over time. Without the natural greenhouse effect moderating global temperatures, life would not be possible on Earth. However, scientists are concerned that increasing levels of greenhouse gases produced by our use of fossil fuels are causing global warming and triggering consequential shifts in weather patterns and climate. This is referred to as the **enhanced greenhouse effect**.

i The greenhouse effect is caused by heat being trapped in the Earth's atmosphere by greenhouse gases, which causes an increase in temperatures at the Earth's surface. As the amount of greenhouse gases in the Earth's atmosphere increases due to human activities, more heat is trapped, which is predicted to cause global changes in climate.

Methane, water vapour, nitrogen oxides and ozone are also greenhouse gases. Methane is 21 times more effective at trapping heat than carbon dioxide. The way in which greenhouse gases restrict heat radiation leaving the Earth is shown in Figure 11.2.3.

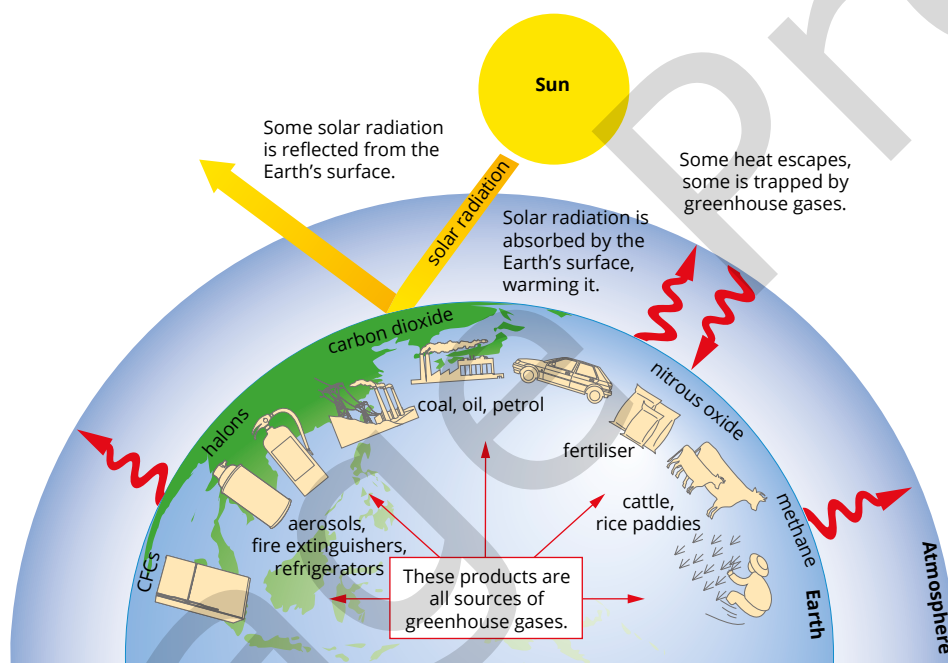


FIGURE 11.2.3 The greenhouse effect. Greenhouse gases help to maintain the temperature at the Earth's surface. Increased quantities of these gases as a result of human activities create an enhanced greenhouse effect.

The graph shown in Figure 11.2.4 on page 238 shows that the Earth is warming. This dramatic rise coincided with the industrial revolution in the late 18th and early 19th centuries. Presently, the concentration of carbon dioxide in the Earth's atmosphere is rising by 0.5% per year. There is overwhelming scientific evidence to conclude that the use of fossil fuels by industrialised countries is the main contributing factor to this increase. Many countries are choosing alternatives to fossil fuels to address these fears.

A further impact on the atmospheric carbon dioxide levels is the continued clearing of forests, particularly in tropical areas. Carbon dioxide is released when trees are burnt or left to decay. In addition, the clearing of the forests will result in less carbon dioxide being removed from the atmosphere by photosynthesis.

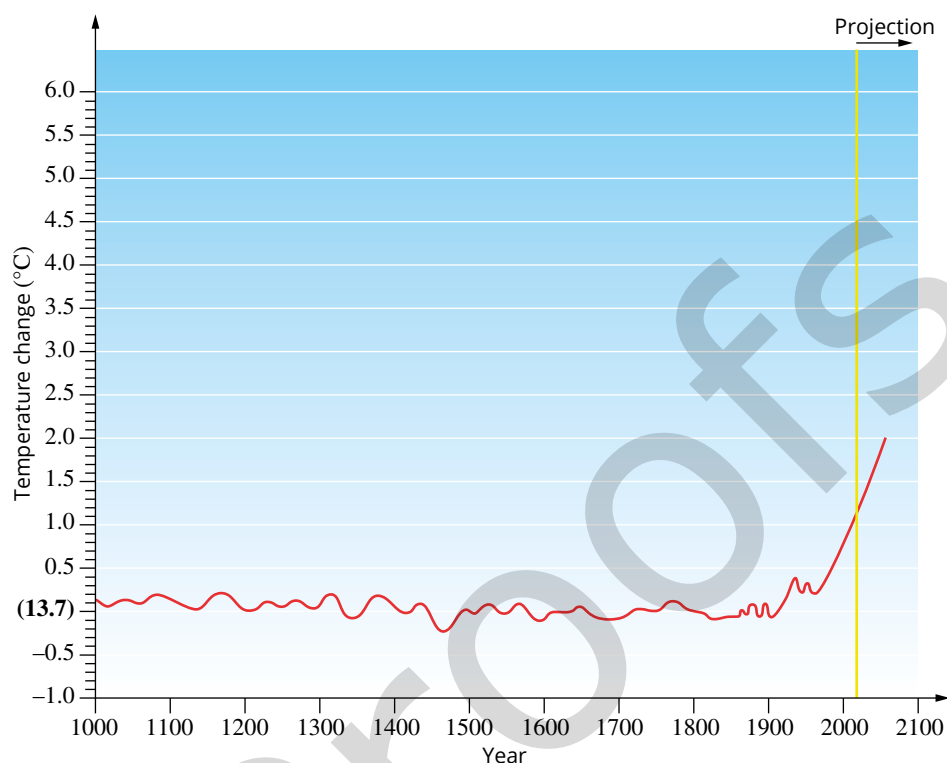


FIGURE 11.2.4 Change in the average surface temperature of the Earth from the year 1000 projected to 2100. Estimates of future temperature increases vary, depending on what assumptions are made.

Global temperatures are on course to rise between 2 and 4 degrees by the end of the century due to global warming. The agriculture in warm regions of the world, including Australia, could feel catastrophic consequences of this temperature rise. Other predictions of the consequences of rapid global warming include:

- Rising sea levels. Sea levels could rise by up to 1.2 m by the year 2100. This is in part due to the melting of polar ice caps.
- Climatic changes. South-eastern Australia is predicted to have increased summer rainfall and decreased winter rainfall. Extreme weather conditions, such as droughts and floods, could be more frequent.
- Ecological changes. These may occur as a result of rising sea levels and climatic change. For example, there could be a loss of natural wetlands.

Each fuel discussed in this chapter produces carbon dioxide when it burns. Table 11.2.1 compares the theoretical mass of carbon dioxide produced from the complete combustion of 1 gram of each fuel and per unit of energy produced.

TABLE 11.2.1 Mass of CO_2 produced from the combustion of 1 gram of fuel and per megajoule of energy produced

Fuel	Mass of CO_2 (g) emitted per gram of fuel	Mass of CO_2 (g) per megajoule of energy produced (approx.)
coal	3.7	93
natural gas	2.8	56
LPG	3.0	65
petrol	3.1	73
ethanol	1.9	72
petrodiesel	3.4	71
biodiesel	2.8	75

CHEMFILE

Cleaner coal

Research is being conducted into ways of reducing carbon dioxide emissions from coal-fired power stations. These include:

- storing carbon dioxide deep underground (geosequestration)
- removing water from brown coal by heating; the volatile hydrocarbons that are driven off are also burnt to produce electricity
- absorbing carbon dioxide into solvents (chemical capture).

There are several methods for converting coal to liquid and gaseous fuels. Flash pyrolysis has been developed by CSIRO for converting coal to oil. In this process, crushed coal is heated to 600°C. The tar that forms is reacted with hydrogen to give a type of crude oil.

Other emissions

Carbon dioxide and water vapour are not the only products formed when fuels are burnt. Fuels may contain elements such as sulfur, which burns to form sulfur dioxide, or the high temperatures of combustion can lead to reactions with nitrogen in air. Table 11.2.2 lists the more common pollutants formed.

TABLE 11.2.2 Pollutants formed from fuel combustion

Emission	Formation	Comments
sulfur dioxide (SO ₂)	sulfur in fuel reacts with oxygen: $S(s) + O_2(g) \rightarrow SO_2(g)$	• causes respiratory problems • leads to the formation of acid rain
nitrogen oxides (NO and NO ₂)	nitrogen in fuel reacts with oxygen, or nitrogen in air reacts due to high temperatures: $N_2(g) + O_2(g) \rightarrow 2NO(g)$ $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$	• causes respiratory problems • leads to the formation of other pollutants and acid rain
ozone (O ₃)	nitrogen oxides react with oxygen at high temperatures	• causes respiratory problems
particulates, such as ash	combustion of impurities in fuel	• irritant • adheres to houses and plants
carbon monoxide	incomplete combustion of fuel	• poisonous gas
organic chemicals, such as methanal and ethanal	reactions of other organic chemicals in fuel	• toxic and carcinogenic compounds

The levels of each pollutant mentioned in Table 11.2.2 vary with the composition of the fuel and the efficiency of the particular combustion process. However, it is possible to generalise.

- Ash from burning is usually more of a problem with coal than other fuels. Ash is produced when coal is burnt without any purification or removal of impurities.
- Sulfur levels are lower in natural gas and LPG than in liquid or solid fossil fuels.
- The molecules in petrol are larger than the molecules in natural gas, LPG and ethanol. As a consequence, the combustion of petrol tends to be less complete. This produces more carbon monoxide and particulates.
- Although biofuels are renewable, they can still produce the same pollutants as fossil fuels when burnt.

Environmental impact of sourcing the fuel

All fuels have to be mined or produced and the environmental impact of sourcing a fuel needs to be considered. Table 11.2.3 shows some of the sources of the fuels described in this chapter.

TABLE 11.2.3 Sources of fuels

Fuel source	Notes
 Biogas collected under domes from a sewage plant	Biogas is often collected from sewage farms and rubbish tips. The gas collection minimises emissions associated with these sites. Because methane is much more effective as a greenhouse gas than carbon dioxide, it is better to collect the methane in biogas and combust it to produce carbon dioxide than to release it directly to the atmosphere.
 Oil rig	Oil rigs, once in place, can operate with low impact on the environment but the issues created when a spill or explosion occurs can be significant.
 Open-cut coal mine	Coal mines can be open cut or underground. Open-cut mines, such as in the Latrobe Valley in Victoria, are damaging to the local environment.
 Harvesting a wheat crop for bioethanol production	Bioethanol is produced from crops. Growing crops requires energy expenditure and the use of resources such as water and fertiliser. Intensive farming can lead to land degradation and erosion. These are larger issues if crops are grown solely to produce ethanol, but of less concern if waste from food crops is used as a raw material. Diverting crops from fuel production could also drive up the cost of food produced from those crops.

CHEMFILE

The Gorgon Gas Project

The \$74.7 billion Gorgon Gas Project (Figure 11.2.5) is located on Barrow Island, a class A nature reserve located 85 km off the Pilbara coast of Western Australia. At this site, liquid natural gas (LNG) is processed from nearby subsea fields for exportation and some domestic use. The project was approved by the Environmental Protection Agency (EPA) in 2007, with many stipulations placed on the project's construction and operation phases. These protections placed limits on the amount of pollutants, like CO₂, mercury, nitrous oxides and hydrogen sulfides that would be produced at the plant.

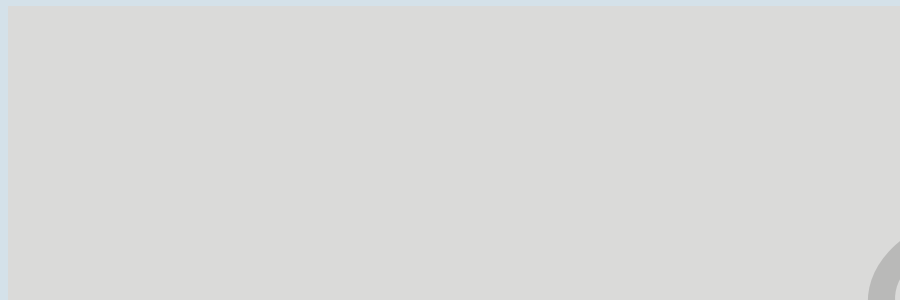


FIGURE 11.2.5 The Gorgon Gas Project, Barrow Island, Western Australia

Additionally, Barrow Island is home for 24 animal species that are native to the island (including the Barrow Island Euro shown in Figure 11.2.6) and others which are classified as vulnerable or endangered, as well as a large variety of flora. This large array of biodiversity is in part due to the isolation of the island, which was separated from mainland Australia about 8000 years ago. This isolation has meant that no introduced predators to Australia have reached the island. As a result, companies that are involved in the project have designed the plant and other amenities to reduce the impact on the environment, as well as keeping a tally on the local animal populations and promoting a sustainable approach to biodiversity in the wider community.



FIGURE 11.2.6 The Gorgon Gas Project has been designed to reduce the environmental impact on Barrow Island's flora and fauna, such as this Barrow Island Euro.

11.2 Review

SUMMARY

- Fuels such as petrol, natural gas, biogas and bioethanol undergo combustion reactions in excess oxygen to form carbon dioxide and water.
- The combustion of fuels can produce a range of other pollutants such as carbon monoxide, sulfur dioxide, nitrogen oxides and particulates.
- Petrol produces more energy per gram than LPG or bioethanol. However, bioethanol and LPG produce less carbon dioxide and particulates in emissions.
- Biofuels offer several environmental advantages: CO_2 is absorbed during the growth of crops used in their production, they can be replenished and they can be produced from material that would have otherwise been waste.
- A shift to large-scale production of biofuels could place a strain on resources and available farmland.

KEY QUESTIONS

- Write a balanced equation for the complete combustion of liquid benzene (C_6H_6).
- Write a balanced equation for the incomplete combustion of ethanol ($\text{C}_2\text{H}_5\text{OH}$) when carbon monoxide is formed.
- Write a balanced equation for the combustion of hydrogen sulfide (H_2S) in excess oxygen.
- How many molecules of oxygen are required to react with each molecule of heptane when it burns completely?
- Describe how the following human activities affect the concentration of carbon dioxide in the atmosphere.
 - burning biofuels
 - burning petrol
 - clearing native forest
 - growing crops to produce biofuels
 - using windpower rather than fossil fuels
 - increased industrialisation
- Explain the effects that the following gases have in the atmosphere.
 - carbon dioxide
 - carbon monoxide
 - methane
 - water vapour
 - sulfur dioxide

11.3 Calculations involving fuels

COMPOSITION OF FUELS

The amount of carbon present in a fuel will influence the amount of energy released by the fuel, the amount of oxygen required for the fuel to combust completely and the amount of carbon dioxide released during the combustion of the fuel. Therefore, we need to know the composition of fuels, or more specifically, the amount of carbon in the fuel, to be able to calculate the energy output of the fuel as well as the carbon dioxide emissions.

Worked example 11.3.1

CALCULATING THE CARBON CONTENT IN COAL

Calculate the mass of carbon present in 1600 tonnes of brown coal that has a carbon content of 75.0%.	
Thinking	Working
Change the percentage to a fraction and multiply by the mass of the substance.	$m(\text{C}) = \frac{75.0}{100} \times 1600 = 1120 \text{ tonnes}$

Worked example: Try yourself 11.3.1

CALCULATING THE CARBON CONTENT IN COAL

Calculate the mass of carbon present in 1300 tonnes of black coal that has a carbon content of 90.0%.

The mass of carbon in compounds used as fuels can be calculated using the formula of the compound and the relative atomic masses (standard atomic weights) of the elements contained in the fuels. Knowing the IUPAC system for naming alkanes will help you in these problems.

Worked example 11.3.2

CALCULATING THE CARBON CONTENT IN PURE COMPOUNDS

Calculate the mass of carbon in 20.0 kg of octane.	
Thinking	Working
Write the formula of the compound and calculate the relative molecular mass.	$M(\text{C}_8\text{H}_{18}) = (8 \times 12.01) + (18 \times 1.008) = 114.2$
Calculate the percentage of carbon in this total.	$\%(\text{C}) = \frac{8 \times 12.01}{114.2} \times 100 = 84.13\%$
Calculate the unknown mass of carbon using this percentage.	$m(\text{C}) = \frac{84.13}{100} \times 20.0 \text{ kg} = 16.8 \text{ kg}$

Worked example: Try yourself 11.3.2

CALCULATING THE CARBON CONTENT IN PURE COMPOUNDS

Calculate the mass of carbon in 25.0 kg of butanol, $\text{C}_4\text{H}_9\text{OH}$.

i The heat of combustion is usually measured at conditions of 298 K and 100 kPa.

i Only fuels that exist as pure substances can have their heat of combustion measured in kJ mol^{-1} .

TABLE 11.3.1 Heats of combustion for some common elements and compounds

Substance	Heat of combustion, ΔH_c (kJ mol^{-1})
methane	-890
ethane	-1560
propane	-2220
butane	-2886
octane	-5450
methanol	-725
ethanol	-1367
hydrogen	-286
carbon (graphite)	-394

ENERGY CONTENT OF FUELS

All substances contain chemical energy. The chemical energy of a substance is referred to as its **energy content**. Fuels are examples of substances with high energy contents.

The **heat of combustion** of a fuel is defined as the enthalpy change that occurs when a specified amount (e.g. 1 g, 1 L, 1 mol) of the fuel burns completely in oxygen. Therefore the energy content of a fuel used in combustion can be represented by the fuel's heat of combustion. Heat of combustion is usually measured at conditions of 298 K (25°C) and 100 kPa, (which means that the water produced should be shown in the liquid state). The heat of combustion can be given the symbol ΔH_c .

Many fuels, including wood, coal and kerosene, are mixtures of chemicals and do not have a specific chemical formula or molar mass. This means their heat of combustion cannot be expressed in kJ mol^{-1} . Therefore, it is measured only as kJ g^{-1} , kJ L^{-1} or MJ/tonne .

The heats of combustion for some common elements and compounds present in fuels are listed in Table 11.3.1. Heat energy is released during combustion, so ΔH_c always has a negative value.

Note that these are theoretical values and, when these fuels are combusted, not all the energy is released as heat energy, and not all of the heat energy liberated can be harnessed for its intended use. However, even if the energy transformation is not 100% efficient, we can still use these values to compare the energy released from the combustion of different compounds.

As Worked Example 11.3.4 shows, you can use the data in Table 2.3.1 to calculate the energy released on combustion of a specified mass of one of the fuels. The energy released when n mol of a fuel burns is given by the equation:

$$\text{Energy} = n \times \Delta H_c$$

Worked example 11.3.4

CALCULATING ENERGY RELEASED BY A SPECIFIED MASS OF A PURE FUEL

Calculate the amount of energy released when 3.60 kg of butane (C_4H_{10}) is burnt in an unlimited supply of oxygen.

Thinking	Working
Calculate the number of moles of the compound using $n = \frac{m}{M}$.	$n(\text{C}_4\text{H}_{10}) = \frac{m}{M}$ $= \frac{3.60 \times 10^3}{58.12}$ $= 61.94$
Multiply the number of moles by the heat of combustion. (Give answer to 3 significant figures.)	$\text{Energy} = n \times \Delta H_c$ $= 61.94 \times 2886$ $= 1.79 \times 10^5 \text{ kJ}$

Worked example: Try yourself 11.3.4

CALCULATING ENERGY RELEASED BY A SPECIFIED MASS OF A PURE FUEL

Calculate the amount of energy released when 5.40 kg of propane (C_3H_8) is burnt in an unlimited supply of oxygen.

Energy content per gram

The energy content of a pure substance can be represented in units of kilojoules per gram, as well as in kilojoules per mole.

For a pure substance, the heat of combustion per gram can be calculated by simply dividing the heat of combustion per mole (kJ mol^{-1}) by the molar mass of the substance.

For example, for ethanol (which is being burnt in a spirit burner in Figure 11.3.1):

Heat of combustion per mole = $-1367 \text{ kJ mol}^{-1}$

Molar mass = $46.068 \text{ g mol}^{-1}$

Heat of combustion per gram = $\frac{-1367}{46.0} = -29.7 \text{ kJ g}^{-1}$

For fuels that are mixtures, approximate values for the heat of combustion per gram are shown in Table 11.3.2.

Energy content per tonne

The quantities of fuel consumed and the amount of energy produced are often so large that the unit megajoules per tonne, MJ/tonne, is a more useful way of expressing the heat of combustion. For example, the Hazelwood power station in Victoria consumed about 13 million tonnes of coal in 1 year prior to being shut down.

$1 \text{ kJ} = 10^{-3} \text{ MJ}$

$1 \text{ g} = 10^{-6} \text{ tonne}$

Therefore, $1 \text{ kJ g}^{-1} = 10^{-3} \text{ MJ} / 10^{-6} \text{ t}$

This can be simplified to $1 \text{ kJ g}^{-1} = 10^3 \text{ MJ/tonne}$.

EXTENSION

Energy density

Liquid fuels, such as petrol, are normally sold by volume rather than by mass. For these fuels, it is often convenient to refer to the heat of combustion per litre. Table 11.3.3 shows some heats of combustion per litre for common liquid fuels. The energy released per litre of fuel is often called the fuel's energy density.

TABLE 11.3.3 Energy densities for some common liquid fuels

Substance	Energy density (kJ L^{-1})
petrol (unleaded 91)	-34 200
kerosene	-36 500
diesel fuel	-38 000
heating oil	-38 500
ethanol	-23 400



FIGURE 11.3.1 Ethanol burning in a spirit burner. Combustion of 1 gram of ethanol releases almost 30 kJ of energy.

TABLE 11.3.2 Approximate heat of combustion values for some fuel mixtures

Substance	Heat of combustion (kJ g^{-1}) (approx.)
wood, dried	-18
peat, dried	-25
brown coal, dried	-30
black coal, dried	-35

i Energy density is the energy released when 1 litre of fuel undergoes complete combustion.

STOICHIOMETRY AND REACTIONS OF FUELS

Due to the growing concerns of fuel combustion on the Earth's environment, it is important to be able to compare the amount of carbon dioxide and other toxic products that will be released from the combustion of different fuel options. Scientists use calculations to predict the amount of carbon dioxide released into the atmosphere when carbon-based fuels are used, or the amount of toxic products that would be produced from fuels that contain impurities such as sulfur.

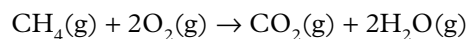
These calculations, which involve the use of the mole concept (Chapter 9), combined with an understanding of chemical equations, are called stoichiometric calculations. In this chapter, these calculations will be used for reactions involving fuels, but stoichiometry is important in all areas of chemistry.

Stoichiometry is the study of ratios of moles of substances. Stoichiometric calculations are based on the law of conservation of mass: *In a chemical reaction, the total mass of all products is equal to the total mass of all reactants.*

Another way of expressing this law is that, in a chemical reaction, atoms are neither created nor destroyed.

Equations and reacting masses

Consider the equation for the reaction that occurs when methane burns in oxygen:



The coefficients used to balance the equations show the ratios between the reactants and products involved in the reaction. The equation indicates that 1 mole of $\text{CH}_4(\text{g})$ reacts with 2 moles of $\text{O}_2(\text{g})$ to form 1 mole of $\text{CO}_2(\text{g})$ and 2 moles of $\text{H}_2\text{O}(\text{g})$. In more general terms, the amount of oxygen used will always be double the amount of methane used, double the amount of carbon dioxide produced and the same as the amount of water vapour produced.

$$\frac{n(\text{O}_2)}{n(\text{CH}_4)} = \frac{2}{1} \quad \frac{n(\text{CO}_2)}{n(\text{O}_2)} = \frac{1}{2} \quad \text{and} \quad \frac{n(\text{H}_2\text{O})}{n(\text{O}_2)} = \frac{2}{2} = 1$$

In general, for calculations you will be given, or you will be able to work out from data, the number of moles of one chemical in the reaction. This is called the 'known'—you know the amount (in moles) of this substance. You will then need to calculate the number of moles of another chemical, the 'unknown'. The ratio(s) of the moles, as shown by the coefficients in the equation, allow you to carry out this calculation.

The relationship used can be written as:

$$n(\text{unknown chemical}) = \frac{\text{coefficient of unknown chemical}}{\text{coefficient of known chemical}} \times n(\text{known chemical})$$

Worked example 11.3.5

USING MOLE RATIOS

How many moles of carbon dioxide are generated when 24 moles of propane are burned completely in oxygen?	
Thinking	Working
Write a balanced equation for the reaction.	$\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$
Note the number of moles of the known substance (in this case propane).	$n(\text{C}_3\text{H}_8) = 24 \text{ mol}$
Write a mole ratio for the chemicals being investigated. $\frac{\text{coefficient of unknown}}{\text{coefficient of known}}$	$\frac{n(\text{CO}_2)}{n(\text{C}_3\text{H}_8)} = \frac{3}{1}$
Calculate the number of moles of the unknown substance using: $n(\text{unknown}) = \text{mole ratio} \times n(\text{known})$	$n(\text{CO}_2) = \frac{3}{1} \times 24$ $= 72 \text{ mol}$

Worked example: Try yourself 11.3.5

USING MOLE RATIOS

How many moles of carbon dioxide are generated when 0.50 moles of butane are burned completely in oxygen?

MASS-MASS STOICHIOMETRY

Calculations usually require us to start and finish with masses rather than moles, as this is how quantities of chemicals are often measured. To calculate the number of moles from the mass of a substance, we can use this relationship:

$$\text{moles} = \frac{\text{mass in grams}}{\text{molar mass}}$$

which is written as

$$n = \frac{m}{M}$$

and to calculate a mass, we can rearrange this relationship to become:

$$m = n \times M$$

Worked example 11.3.6

SOLVING MASS–MASS STOICHIOMETRIC PROBLEMS

Calculate the mass of carbon dioxide produced when 5400 g of propane, C_3H_8 , is burned completely in oxygen.	
Thinking	Working
Write a balanced equation for the reaction.	$\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$
Calculate the number of moles of the known substance (in this case propane) using: $n = \frac{m}{M}$	$n(\text{C}_3\text{H}_8) = \frac{5400}{44.094}$ $= 122.5 \text{ mol}$
Find the mole ratio: $\frac{\text{coefficient of unknown}}{\text{coefficient of known}}$	$\frac{n(\text{CO}_2)}{n(\text{C}_3\text{H}_8)} = \frac{3}{1}$
Calculate the number of moles of the unknown (in this case carbon dioxide) substance using: $n(\text{unknown}) = \text{mole ratio} \times n(\text{known})$	$n(\text{CO}_2) = \frac{3}{1} \times 122.5$ $= 367.4 \text{ mol}$
Calculate the mass of the unknown substance using: $m = n \times M$	$m(\text{CO}_2) = 367.4 \times 44.01$ $= 16\,169 \text{ g}$ $\approx 16.2 \text{ kg}$

Worked example: Try yourself 11.3.6

SOLVING MASS–MASS STOICHIOMETRIC PROBLEMS

Calculate the mass of carbon dioxide produced when 3600 g of butane, C_4H_{10} , is burned completely in oxygen. (The molar mass of butane is 58.12 g mol^{-1} .)

CHOOSING A FUEL

A wide range of fuels is used for many different purposes. As we have seen in this chapter, the factors that need to be considered when selecting a fuel for a particular purpose include:

- energy released per unit mass or unit volume
- availability and cost of fuel
- ease of transport
- hazards to people and the environment associated with its use and its waste products
- social, economic, cultural and political values that can affect the choice of a fuel.

We can find many examples of these factors on fuel choice. Liquid hydrogen was chosen for the *Saturn* rockets used in the US space program because it provides large quantities of energy per gram. As space rockets need to carry their own fuel, this was a crucial factor.

One of the key concerns for fuel choice for electricity production is efficiency. The term **energy efficiency** is used to describe the percentage of energy from a source that is converted to useful energy. For example, if the efficiency of a solar panel on a roof (Figure 11.3.2) is listed as 17%, it means that 17% of the energy



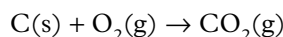
FIGURE 11.3.2 Commercial solar cells convert solar energy to electrical energy with an efficiency of 12–18%.

arriving on the panel from the Sun is transformed to electrical energy. The other 83% is converted to other forms of energy. The largest proportion of the Sun's energy reaching the solar cells is converted into heat energy that simply increases the temperature of the cells.

In Australia, electrical energy is produced from several different fuels.

Electricity from coal

The combustion of coal generates over three-quarters of Australia's electricity. Rather than transport coal to every factory, office and household, the chemical energy is converted to electrical energy at a power station. Electricity is transmitted easily from the power station by metal cables and wires to other regions. The reaction occurring when coal burns can be written as:



The energy released from the combustion of coal is about 32 kJ g^{-1} .

A number of energy transformations occur in a coal-fired power station.

- Coal is burnt—chemical energy in coal is converted to thermal energy.
- Heat from the burning coal is used to boil water—thermal energy from the burning coal becomes thermal energy in steam.
- Steam is passed through a turbine—thermal energy in the steam becomes mechanical energy as the turbine spins. (This is the least efficient energy transformation in the sequence.)
- Electricity is produced from a generator that is driven by the turbine—mechanical energy is converted to electrical energy.

Figure 11.3.3 illustrates how the thermal energy released by the coal is converted to electrical energy.

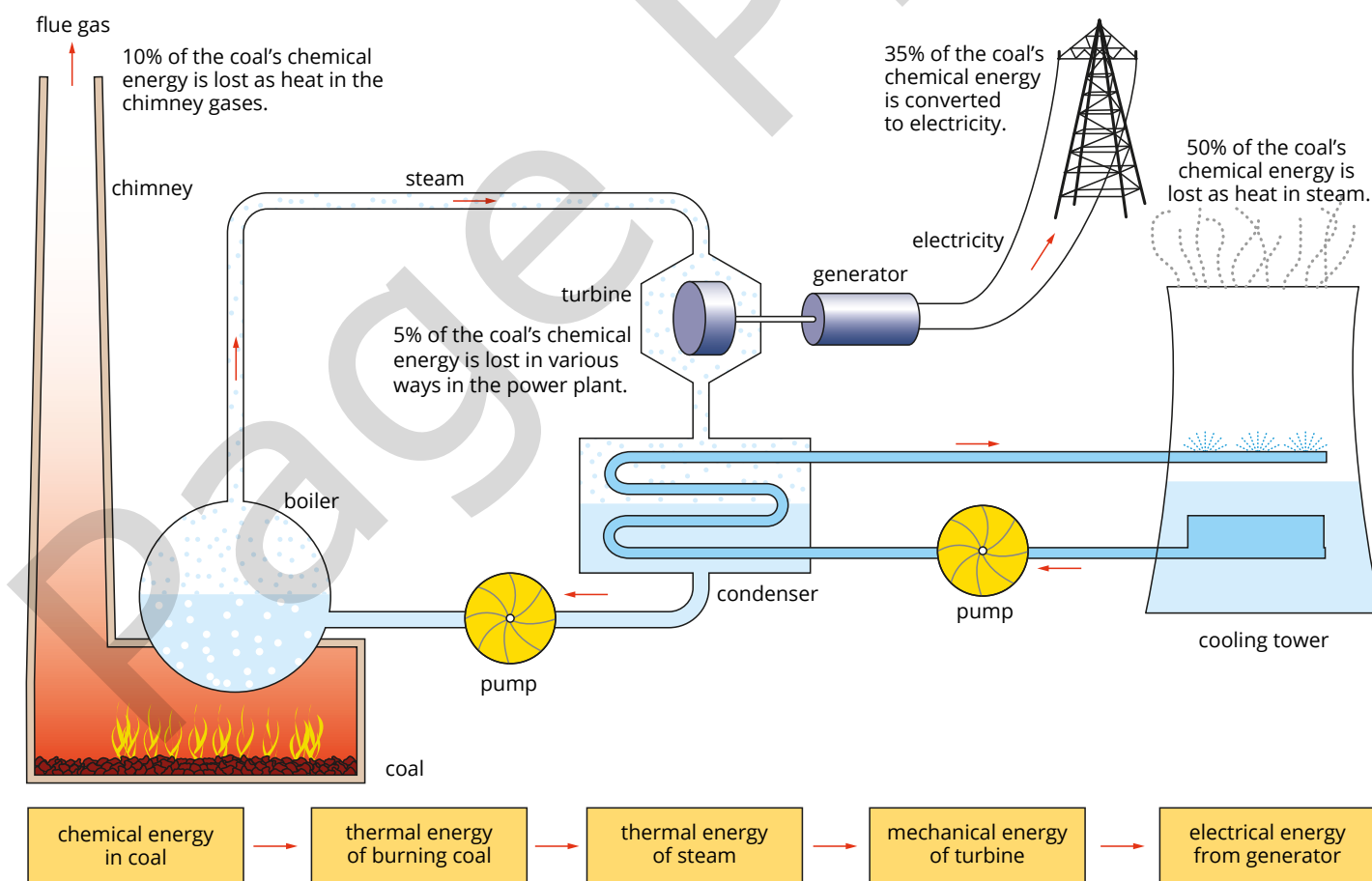


FIGURE 11.3.3 A representation of the process of burning coal to produce electricity. The chemical energy in the coal undergoes several transformations before electricity is produced.

The overall efficiency of a coal-fired power station is 30–40%. The combustion of brown coal is usually at the lower end of this efficiency range. Energy is lost during each step of the process, mainly as heat.

Electricity from natural gas

Natural gas is also used in a number of locations in Western Australia, including Pinjar and Kwinana, to generate electricity for both residential and industrial purposes. In a gas-fired power plant, methane and other small alkanes are burnt to release energy. As shown in Figure 11.3.4, the hot gases produced by combustion cause air to expand in a combustion turbine to generate electrical energy. This is a simpler process than in a coal-fired plant where the thermal energy is used to produce steam.

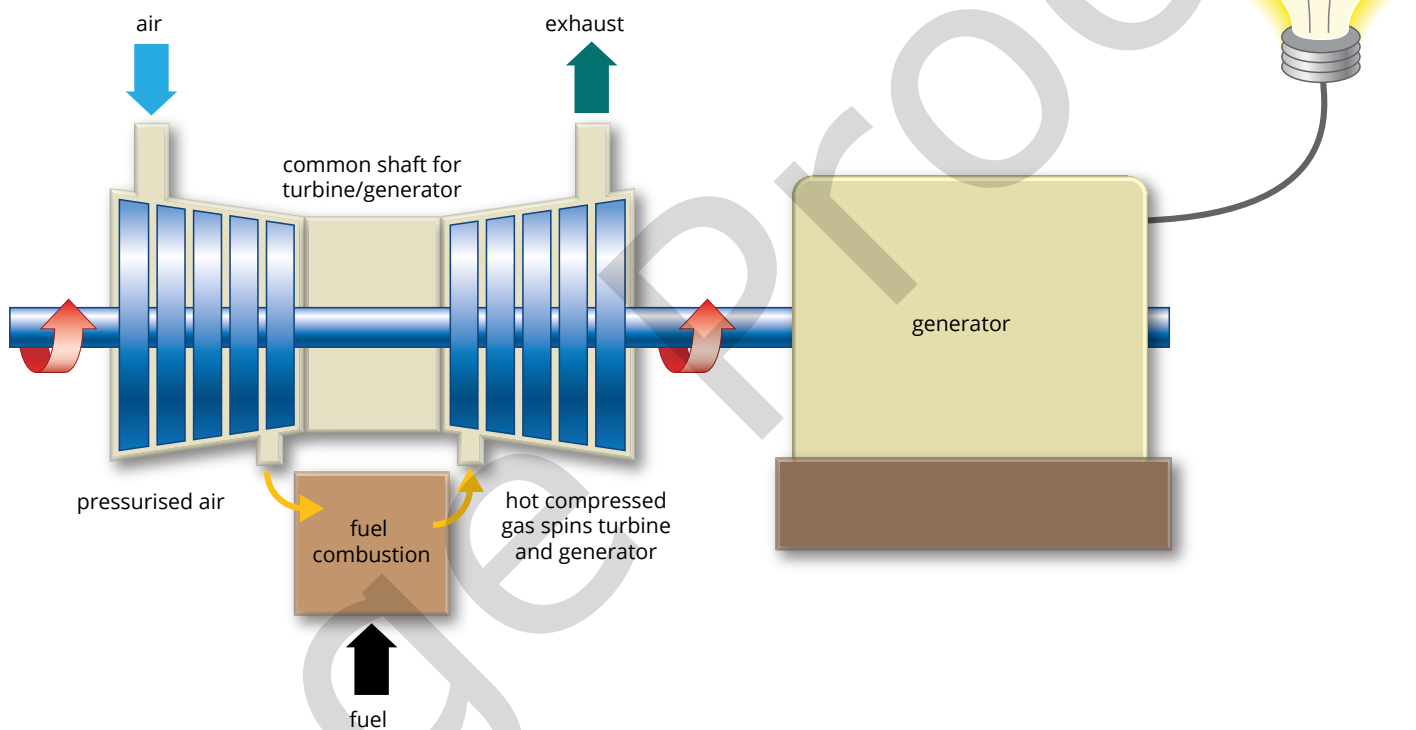
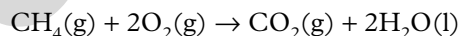


FIGURE 11.3.4 In a gas-fired power plant, the hot gases produced expand air in a combustion turbine to generate electricity.

The composition of natural gas varies but the main combustion reaction involves methane. The equation is:



The combustion of 1.00 mole of methane releases 890 kJ of energy, equivalent to 55.6 kJ g⁻¹. This is a significantly higher value than that of coal.

A gas-fired plant is more efficient than a coal-fired power station, reaching efficiencies just over 40%. Gas-fired plants also emit less carbon dioxide and particulate matter (small solid particles of solid combustion products) per unit of energy released. An added advantage of gas-fired plants is that the output can be varied at short notice. This allows the operators to adjust to the fluctuating power usage of consumers. Most Australian states have gas-fired plants but many of the plants are small-scale ones. Coal seam gas is the source of methane used in some states.

Electricity from biogas

Biogas is a renewable fuel that can be used to generate electricity, usually in small-scale electricity generators rather than large power plants. These smaller generators are often located at the site where the biogas is produced. For example, sewage works commonly burn biogas produced in a generator to supply some of their power needs.

The main reaction occurring in the combustion of biogas is the same reaction of methane burning in a gas-fired power station. The energy released per gram of biogas is less than that of natural gas because the methane content in biogas is significantly lower.

Berrybank Farm near Ballarat, Victoria is an example of the innovative use of biogas. Over \$2 million has been spent on building infrastructure to collect the manure from 20 000 pigs (Figure 11.3.5a). The manure is fed into a digester that produces two useful products: biogas and fertiliser. The biogas is used to fire generators, like the one shown in Figure 11.3.5b, that produce an estimated \$180 000 of electricity annually.

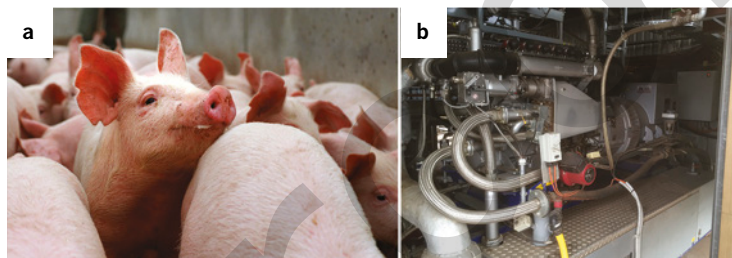


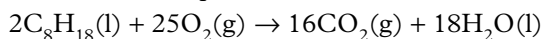
FIGURE 11.3.5 (a) Some of the pigs on the Berrybank Farm near Ballarat; (b) one of the generators that uses the biogas fuel

FUEL FOR TRANSPORT

Crude oil is the source of most of the fuel we use for transport. Crude oil is a mix of alkanes. The alkanes in crude oil are separated into a series of fractions (parts) by fractional distillation. Some of these fractions are important fuels, such as liquid petroleum gas (LPG), petrol, kerosene and petrodiesel. LPG can also be separated from natural gas.

Petrol

Perhaps the most important of all hydrocarbon combustion reactions are those that occur when petrol is burnt. Petrol is a mixture of hydrocarbons, including octane, and the combustion reactions of these chemicals power most of Australia's 17.6 million motor vehicles. The equation for the combustion of octane is:



The combustion of 1.00 mole of octane releases 5450 kJ of energy, equivalent to 47.8 kJ g^{-1} . Combustion occurs in the cylinder of a car engine. The hot gases formed push the piston in the engine, enabling the car to move. A typical piston is shown in Figure 11.3.6.

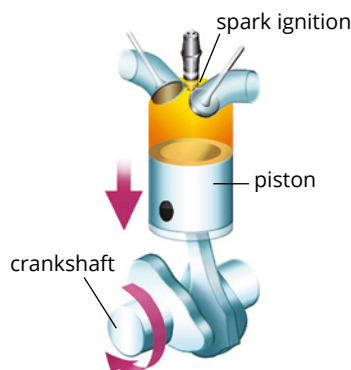


FIGURE 11.3.6 The combustion of octane (C_8H_{18}) and the other hydrocarbons in petrol pushes the pistons in internal combustion engines.

The efficiency of a petrol engine in a new car can be as high as 25%. The operation of a piston in a car engine can be seen in Figure 11.3.7.

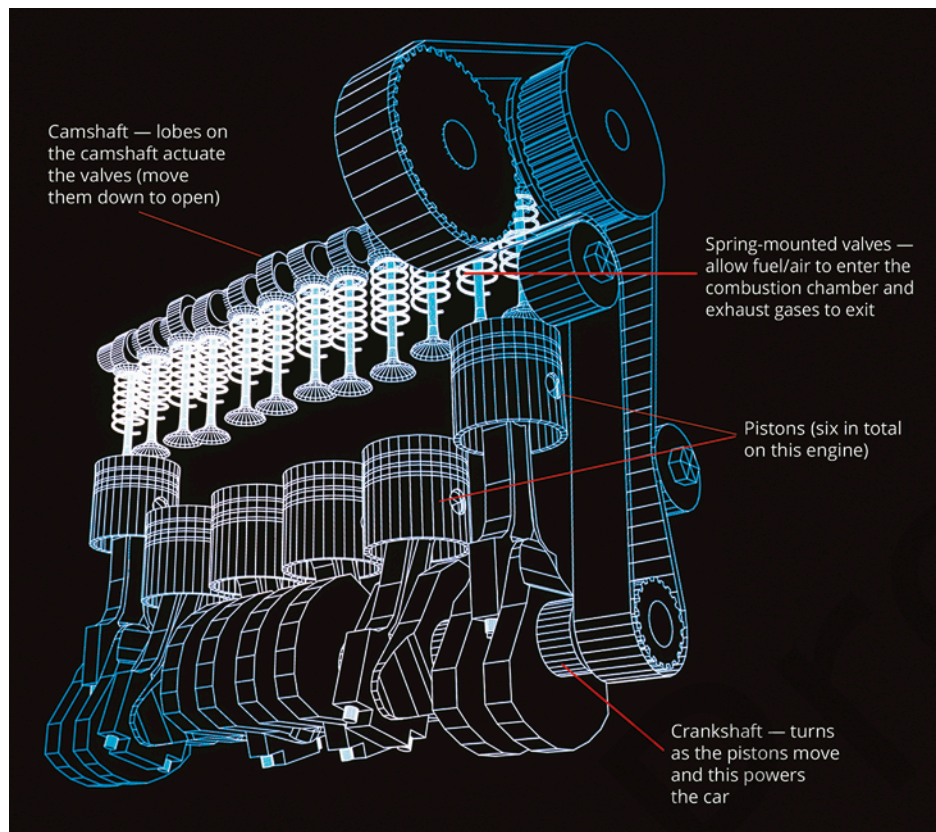
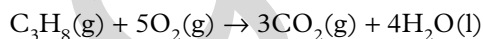


FIGURE 11.3.7 In a standard engine, fuel and air are mixed in cylinders and then compressed, and the mixture is ignited. The explosion drives the piston down, which turns the crankshaft and powers the car. The flow of fuel and exhaust gases is controlled by spring-mounted valves shown in the centre.

Liquid petroleum gas

Liquid petroleum gas (LPG) can also be used in cars. Most of the vehicles that use LPG as a fuel have a standard petrol engine with a fuel tank and fuel injection system modified to suit a gaseous fuel. The equation for the combustion of propane, a major component of LPG, is:



The combustion of 1.00 mole of propane releases 2220 kJ of energy, equivalent to 50.5 kJ g^{-1} .

In Australia, LPG is a significantly cheaper fuel than petrol, yet its popularity is still limited. Some of the reasons for this are:

- most new vehicles are designed to run on petrol; therefore, the owner has to pay around \$2000 for a conversion
- the LPG fuel tank takes up boot space
- there are fears that LPG cylinders might explode if the vehicle crashes
- the prices of fuels fluctuate, so often it is difficult to do meaningful price comparisons.

Bioethanol

Australia imports over 90% of its fuel requirements. This reliance on other countries, combined with concerns over the greenhouse emissions of fossil fuels, has sparked interest in the production of the renewable biofuels such as bioethanol.

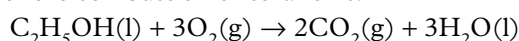
Bioethanol can be produced from crops such as sugar cane. However, sugar cane is also needed for sugar production so there are limits to the amounts of bioethanol that can be produced in this way. Instead, researchers are trialling less valuable sources of sugar and starch for bioethanol production.

The Manildra plant at Nowra in New South Wales, shown in Figure 11.3.8, is one of Australia's largest ethanol refineries. At this plant, flour and starch are produced from wheat and sold for use in food manufacture. The waste that remains still contains high levels of starch, which is converted to ethanol.



FIGURE 11.3.8 Ethanol refinery in Manildra at Nowra, New South Wales

The equation for the combustion of ethanol is:



The combustion of 1.00 mole of ethanol releases 1367 kJ of energy, equivalent to 29.7 kJ g^{-1} . As Table 11.3.4 shows, the energy content of ethanol is about 62% that of petrol, so a larger mass (and volume) of ethanol is required to provide the same amount of energy. At a simple level, the lower energy content of ethanol can be regarded as the result of the carbon atoms in an ethanol molecule being partly oxidised ('partly burnt'). This is due to the presence of oxygen in the ethanol molecule.

TABLE 11.3.4 Energy content and energy density of vehicle fuels

Fuel	Energy content (kJ g^{-1})	Energy density (kJ L^{-1})
methane	55.6	23 500 (liquefied)
propane (LPG component)	50.5	29 400 (liquefied)
butane (LPG component)	49.6	29 800 (liquefied)
octane (petrol fraction)	47.8	33 400
ethanol	29.7	23 400
petrodiesel	48.1	40 000
biodiesel	37.8	36 000

11.3 Review

SUMMARY

- Heat of combustion, ΔH_c , indicates the maximum amount of energy that can be released when a specified amount of fuel undergoes complete combustion. Common units are kJ mol^{-1} , kJ g^{-1} and MJ/tonne .
- The amount of energy released by different fuels can be compared by referring to heats of combustion.
- For a pure fuel with a heat of combustion, ΔH_c , measured in kJ mol^{-1} , the energy released when n mol of the fuel burns is given by the equation:

$$\text{Energy} = n \times \Delta H_c$$

- For a fuel that is not pure and a heat of combustion, ΔH_c , measured in kJ g^{-1} , the energy released when m grams of the fuel burns is given by the equation:

$$\text{Energy} = m \times \Delta H_c$$
- The amount of carbon present in a fuel will influence the amount of oxygen needed for combustion and thus the amount of carbon dioxide produced
- Stoichiometry is the study of ratios of moles of substances, with stoichiometric calculations being based on the law of conservation of mass.
- Using stoichiometric ratios you can predict the amount of CO_2 and other harmful emissions created from the combustion of different fuels.
- The combustion reactions of fuels are used to produce electricity and to power vehicles.
- In a coal-fired power station, thermal energy from coal creates steam that is used to turn a turbine and generate electricity. The efficiency of this process is between 30 and 40%.
- Electricity can also be produced from the combustion of natural gas or biogas.
- Petrol engines can use petrol or petrol blended with ethanol. Engines can also be modified to run on LPG.
- Biodiesel, which is sourced from biomass, provides an alternative to petrodiesel.
- Table 11.3.5 compares the advantages and disadvantages of some fuels.

TABLE 11.3.5 Advantages and disadvantages of fuels described in this chapter

Fuel	Advantages	Disadvantages
coal	- large reserves - less easily transported than liquid or gaseous fuels - relatively high energy content	- non-renewable - high level of emissions
natural gas	- more efficient than coal for electricity production - easy to transport through pipes - relatively high energy content	- non-renewable - limited reserves - polluting (but less than coal and petrol)
biogas	- renewable - made from waste - reduces waste disposal - low running costs CO_2 absorbed during photosynthesis	- low energy content - supply of waste raw materials limited
petrol	- high energy content - ease of transport	- non-renewable - polluting (but less than coal) - limited reserves
LPG	- low cost - easily separated from natural gas - relatively high energy content; fewer particulates produced than petrol	- non-renewable - polluting (but less than petrol)
bioethanol	- renewable - can be made from waste CO_2 absorbed during photosynthesis - burns smoothly - fewer particulates produced than petrol	- limited supply of raw materials from which to produce it - lower energy content than petrol - may require use of farmland otherwise used for food production
Biodiesel	- renewable - can be made from biomass - no modifications to vehicles required	- limited supply of raw materials for production - lower energy content than petrodiesel

Review (Continued)

KEY QUESTIONS

- How many moles of water are produced when 50 moles of butanol ($\text{C}_4\text{H}_9\text{OH}$) are burned in an unlimited supply of oxygen?
- Information about two hydrocarbon fuels, propane and octane, is given in the table below.

Characteristic	Propane (C_3H_8)	Octane (C_8H_{18})
heat of combustion (kJ mol^{-1})	-2220	-5450
molar mass (g mol^{-1})	44.0	114.0

Calculate the heat of combustion of each fuel in kJ g^{-1} and use your answer to state which fuel produces more energy per kilogram.

- Calculate the mass of carbon in 1.00 kg of butane (C_4H_{10}).
- Using the information in Tables 11.3.1 on page 244 and 11.3.2 on page 245, calculate the amount of energy released when the following amounts of each fuel undergo complete combustion.
 - 250 g of methane
 - 9.64 kg of propane
 - 403 kg of ethanol
 - 573 t of dried brown coal
- Octane (C_8H_{18}) is a component of petrol. It burns in oxygen to produce carbon dioxide and water. Energy is released during this reaction. The equation for this reaction is:
$$2\text{C}_8\text{H}_{18}(\text{g}) + 25\text{O}_2(\text{g}) \rightarrow 16\text{CO}_2(\text{g}) + 18\text{H}_2\text{O}(\text{g})$$
 - Calculate the mass of oxygen required to react with 200 g of octane.
 - Calculate the mass of carbon dioxide produced in part a.
- A combustion reaction occurred in which 1.00 kg of butane (C_4H_{10}) reacted with excess oxygen gas.
 - Calculate the mass of oxygen, in kg, required for the complete combustion of this amount of butane.
 - Calculate the mass of carbon dioxide, in kg, produced in this reaction.
- Classify each of the following as an advantage or a disadvantage of the use of bioethanol compared to petrol as an energy source.
 - less CO_2 impact overall
 - lower energy content (kJ g^{-1})
 - can be produced from waste products
 - renewable resource
 - greater amount of CO_2 emitted to travel a set distance
- The water content of brown coal is 60–70%. What implications does this high water content have for the energy released from burning the coal?
 - What pre-treatment could raise the energy content per gram of brown coal consumed in a power station?
 - If some electricity produced from burning coal was used to reduce its water content, then how could this energy cost be minimised?
 - What impact do impurities such as sulfur have on generating electricity from brown coal?

Chapter review

KEY TERMS

acidification
biodiesel
bioethanol
biofuels
biogas
carbon neutral
combustion
complete combustion
energy content

energy density
energy efficiency
enhanced greenhouse effect
enzyme
fermentation
fracking
fuel
greenhouse effect

greenhouse gas
heat of combustion
incomplete combustion
liquefied petroleum gas (LPG)
natural gas
non-renewable
petrodiesel
photosynthesis

renewable
SI units
stoichiometry
sustainability

Types of fuels

- The world has become very dependent on the products of the petrochemical industry, but the raw materials of crude oil and natural gas are likely to be virtually exhausted by 2100. Assuming that current production remains unchanged and no alternative sources are available, suggest the impact of the lack of raw materials on our lifestyle.
- Refer to Figure 11.1.4 on page xx to answer the following questions.
 - In the year 2000, what type of fuel provided the most energy on a global scale?
 - Rank the different fuels in order from most to least amount of energy expected to be produced in 2020.
- Why are fossil fuels considered to be non-renewable?
- Refer to Figure 11.1.5 on page xx to answer the following questions.
 - What type of coal takes the longest amount of time to form?
 - Of peat, brown coal and black coal, which releases the least amount of heat energy?
 - Which type of coal is a better quality fuel?
- The following energy sources are used across the world: coal, bioethanol, biodiesel, natural gas. Which one is likely to run out the fastest?
- Explain why bioethanol is sometimes described as a 'carbon neutral' fuel. Use chemical equations for photosynthesis, fermentation and combustion to support your answer.
- Use the terms 'methane', 'oxygen', 'bacteria' and 'carbon dioxide' to explain the formation and composition of biogas.

Combustion reactions

- Write a balanced equation for the complete combustion of butanol (C_4H_9OH), a commonly used biofuel.
- Write a balanced equation for the incomplete combustion of butane (C_4H_{10}) where carbon monoxide is formed.

- The fact that global warming is taking place is now generally accepted. Conduct some research using the internet to discover some of the consequences of global warming. Give one example each of the effect on:
 - the polar ice caps
 - changing weather patterns
 - crop production
 - extinction of plant or animal species.

Calculations involving fuels

- Calculate the energy released when 5.00 kg of methane (CH_4) is burnt in an unlimited supply of oxygen. The heat of combustion of methane is -890 kJ mol^{-1} . (Give your answer in megajoules, MJ.)
- Propane burns in oxygen according to the equation:
$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$
6.70 g of propane was burned in excess oxygen.
 - What mass of carbon dioxide would be produced?
 - What mass of oxygen would be consumed in the reaction?
 - What mass of water would be produced?
- Ethanol can be used as a substitute for petrol. The heat of combustion for ethanol, CH_3CH_2OH , is $-1367 \text{ kJ mol}^{-1}$.
 - Write a balanced equation to represent the complete combustion of ethanol in air.
 - Calculate the energy released when 5.00 g of ethanol burns completely in air.
 - Calculate the mass of carbon dioxide released in this process.
- Methane and propane are used as alternative transport fuels. The heats of combustion of methane and propane are -890 kJ mol^{-1} and $-2220 \text{ kJ mol}^{-1}$ respectively. Calculate the mass, in kg, of each gas required to release 100 000 kJ of heat energy and calculate the mass of carbon dioxide gas produced in the process. Compare the differences between the two fuels.

- 15 a** Explain what 'E10 petrol' is.
b How does the introduction of E10 help with the potential shortage of crude oil?
- 16** What are some of the ways in which energy is lost, leading to a reduced efficiency, in a coal-fired power station?

Connecting the main ideas

- 17** In a steelworks, carbon monoxide present in the exhaust gases of the blast furnace can be used as a fuel elsewhere in the plant. It reacts according to the equation:
- $$\text{CO(g)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} \quad \Delta H = -283 \text{ kJ mol}^{-1}$$
- a** Which has the greater total enthalpy: 1 mol of CO(g) and 0.5 mol of O₂(g) or 1 mol of CO₂(g)?
b Write the value of ΔH for the following equations:
i $2\text{CO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{CO}_2\text{(g)}$
ii $2\text{CO}_2\text{(g)} \rightarrow 2\text{CO(g)} + \text{O}_2\text{(g)}$
- 18** In a car engine, small amounts of the petrol are vaporised, enter the cylinder and are ignited by a spark. If 12.0 mg of octane is released into the cylinder, what mass of oxygen would be required to be present for complete combustion?
- 19** An indoor gas heater burns propane, (C₃H₈) at a rate of 12.7 grams per minute.
a Calculate the minimum amount of oxygen per minute that needs to be available for the complete combustion of the propane.
b What will happen if the flow of oxygen per minute is insufficient for the complete combustion of the propane?
- 20** Ethanol is produced industrially by reacting ethene with water using a phosphoric acid catalyst at 300°C:
- $$\text{C}_2\text{H}_4\text{(g)} + \text{H}_2\text{O(g)} \rightarrow \text{CH}_3\text{CH}_2\text{OH(g)}$$
- a** Explain whether ethanol produced by this method is a biochemical fuel.
b Describe how ethanol, which is classified as a biochemical fuel, could be produced.

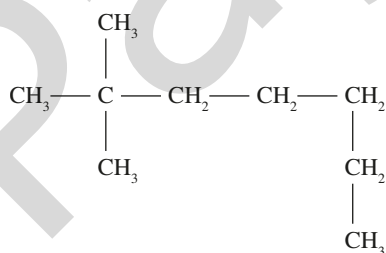
- 21** Methane, ethanol, octane and biodiesel are used to power vehicles.
a List the fuels in order of their energy content per gram (from highest to lowest). You may need to refer to Table 11.3.4 on page xx.
b The emissions of carbon dioxide per gram from the combustion of bioethanol are less than that from octane (petrol). However, a car using bioethanol produces more carbon dioxide when driving the same distance as a car using octane. Which one of the follow could be the best explanation for this difference?
A Bioethanol is more efficient than octane.
B The energy content of bioethanol is less than that of octane.
C The temperature of the engine favours the combustion of octane.
D More energy transformations are required in the combustion of bioethanol.
- 22** Conduct some research on the internet to find out which nations are the top ten consumers of energy.
- 23** The 2015 Paris Agreement was an international response to global warming in which all nations were asked to commit to keeping the global average temperature rise to below 2°C, through reductions in greenhouse gas emissions.
a What impact would adoption of the Paris Agreement have on Australia?
b Discuss the role that biofuels could play in helping Australia meet its target for reducing greenhouse gas emissions.

UNIT 1 • CHEMICAL FUNDAMENTALS: STRUCTURE, PROPERTIES AND REACTIONS

REVIEW QUESTIONS

Section 1: Multiple choice

- Which one of the following statements best describes an element in group 17 of the periodic table?
 - an ion with a charge of negative one
 - an element that gains one electron to achieve a full valence shell
 - an element that can form covalent network solids
 - a noble gas
- Which one of the following is the correct electron configuration of oxygen?
 - 2,8,8
 - 2,2,6
 - 2,6
 - 2,8
- An element 'X' forms a chloride with the formula XCl_3 . Which one cannot be element X?
 - Al
 - Fe
 - N
 - Sr
- Which one of the following solids is classified as a molecular solid?
 - silicon (Si)
 - alumina (Al_2O_3)
 - bronze (a mixture of Cu and Sn)
 - dry ice (solid CO_2)
- In which one of the following solids are both ionic and covalent bonds present?
 - iodine
 - lead iodide
 - ammonium chloride
 - hydrogen iodide
- What is the correct IUPAC name to give to the substance represented by the structure below?



- 2,2-dimethylheptane
- 2,2,4-trimethylhexane
- 1-ethyl-1,3,3-trimethylbutane
- octane

- Which sequence of steps is most likely to be used to separate a pure sample of salt from a mixture of salt and charcoal?
 - distillation, evaporation, filtration, crystallisation
 - filtration, crystallisation, dissolution, evaporation
 - dissolving, filtration, evaporation, crystallisation
 - dissolving, evaporation, filtration, distillation
- Which expression shows the mass of a nitrogen molecule?
 - $\frac{28.0}{6.022 \times 10^{23}}$
 - 14.0g
 - $\frac{6.022 \times 10^{23}}{28.0}$
 - 28.0g
- Which one of the following are all non-renewable sources of energy?
 - natural gas, fuel oil, hydroelectric power
 - coal, biomass to produce ethanol, oil
 - crude oil, wood, natural gas
 - natural gas, coal, bottled gas (LPG)
- From the table below, identify two elements that are isotopes. (Select A, B, C or D.)

Element	Number of protons	Number of electrons	Number of neutrons
W	20	21	21
X	19	18	19
Y	19	21	19
Z	20	19	20

- elements X and W
 - elements X and Y
 - elements W and Z
 - elements Y and W
- $\text{C}_6\text{H}_{14} + \text{Br}_2 + \text{UV light} \rightarrow \text{C}_6\text{H}_{13}\text{Br} + \text{HBr}$
The above reaction is an example of:
 - addition.
 - substitution.
 - combustion.
 - sublimation.
 - A solution may be best described as:
 - a pure substance of constant composition.
 - a homogeneous mixture of uniform composition.
 - a substance that can be purified by filtration.
 - a heterogeneous mixture of variable composition.

UNIT 1 • REVIEW

Section 2: Short answer

1 a Write the formula for the following compounds.

- i gold(III) sulfide
- ii strontium hydroxide
- iii ammonium phosphate

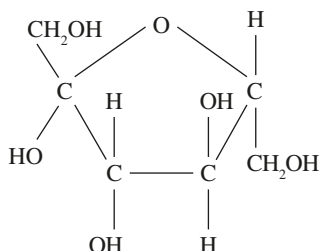
b Write the name for the following substances.

- i $\text{Fe}_2(\text{SO}_4)_3$
- ii P_2O_5

2 One type of sugar found in fruit is fructose.

The molecular formula for fructose is $\text{C}_6\text{H}_{12}\text{O}_6$.

The structural formula is:



- a What is the mass of **1 mole** of fructose? Provide your answer to 4 significant figures.
- b What is the mass of **1 molecule** of fructose? Provide your answer to 2 significant figures.
- c Why do fructose molecules attract each other?
- d What type of chemical bonds exist between the fructose molecules in the solid sugar?

3 a Draw the **electron dot structure** for the following molecules including all bonding and non-bonding electrons.

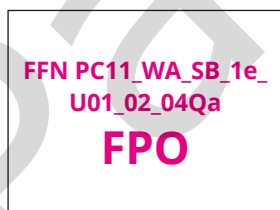
- i water (H_2O)
- ii dichlorodifluoromethane (CCl_2F_2)

b Draw the **structural formula** for the following molecules.

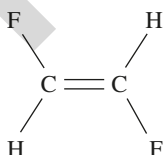
- i *cis*-dibromoethene ($\text{C}_2\text{H}_2\text{Br}_2$)
- ii carbon disulfide (CS_2)

4 a Using IUPAC nomenclature name the following organic compounds.

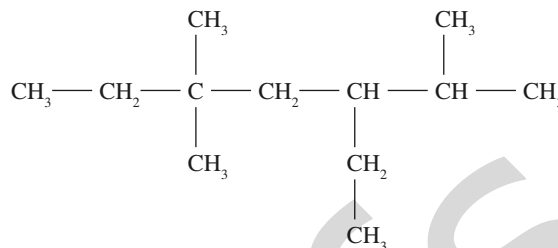
i



ii



iii



b State which of the compounds from part a (i, ii and/or iii) would undergo a substitution reaction with chlorine gas, by circling the correct answer/s.

c Write the **molecular formula** for compound (iii) in part a.

5 a Calculate the percentage of water in aluminium nitrate hexahydrate, $\text{Al}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$. Give your answer to 3 significant figures.

b Calculate how many moles are in 1.00 kg of aluminium nitrate hexahydrate.

6 Copy and complete the table.

Reaction	$\Delta H_{\text{reaction}}$	Exothermic or endothermic
$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g}) + 287 \text{ kJ mol}^{-1}$		
$\text{H}_2\text{S}(\text{g}) + 90 \text{ kJ mol}^{-1} \rightarrow \text{H}_2(\text{g}) + \text{S}(\text{s})$		

Section 3: Extended answer

1 Materials are pure substances with distinct measurable and observable properties. These properties can be related to the types of chemical bonds between the atoms, ions and molecules from which matter is composed.

Electrical conductivity depends on whether these bonds are metallic, ionic, covalent network or covalent molecular.

In terms of chemical bonding, explain how the following solids are electrical conductors or insulators (non-conductors). Labelled diagrams can be used in your answers.

- a copper (Cu)
- b sodium chloride (NaCl)
- c water (ice, H_2O)
- d graphite (C)
- e diamond (C)

- 2** Butane (C_4H_{10}) is a gaseous hydrocarbon fuel that is often used, among many other things, as an energy source for portable stoves.
- a** Draw the structural formula of butane.
 - b** There is another hydrocarbon that has the formula C_4H_{10} .
Draw the structural formula for that isomer and name the compound.
 - c** Using the molecular formula (C_4H_{10}) write the balanced equation for the complete combustion of butane in air.
The enthalpy of reaction for the above equation is $-5750 \text{ kJ mol}^{-1}$.
 - d** Is the combustion of butane exothermic or endothermic?
 - e** Rewrite your equation including the heat (enthalpy) of reaction as a reactant or product.
 - f** Calculate the mass of carbon dioxide produced for every 1.00 kg of butane burned.
Butane undergoes a slow substitution reaction with gaseous chlorine.
 - g** Using molecular or structural formulae, write the equation for the reaction between one mole of butane and one mole of fluorine.
 - h** Write the IUPAC name of the organic product.
An alkene with four carbon atoms is but-1-ene.
 - i** Draw the structural formula for but-1-ene.
 - j** Using structural formulae, write the equation for the reaction between but-1-ene and bromine.
 - k** Name the organic product formed in the reaction in part j.



LOW-RES